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Classification of elements and Periodicity in Properties - Periodic Table

1. The correct increasing order of the ionic radii is

(1) $Cl^- < Ca^{2+} < K^+ < S^{2-}$

(2) $K^+ < S^{2-} < Ca^{2+} < Cl^-$

(3) $S^{2-} < Cl^- < Ca^{2+} < K^+$

(4) $Ca^{2+} < K^+ < Cl^- < S^{2-}$

CPT243

2. Given below are two statements:

Statement I : The decrease in first ionization enthalpy from B to Al is much larger than that from Al to Ga.

Statement II : The d orbitals in Ga are completely filled.

In the light of the above statements, choose the most appropriate answer from the options given below

(1) Statement I is incorrect but statement II is correct.

(2) Both the statements I and II are correct

(3) Statement I is correct but statement II is incorrect

(4) Both the statements I and II are incorrect

CPT244

3. Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R)

Assertion (A) : The first ionization enthalpy of 3d series elements is more than that of group 2 metals

Reason (R) : In 3d series of elements successive filling of d-orbitals takes place.

In the light of the above statements, choose the correct answer from the options given below :

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is false but (R) is true

(4) (A) is true but (R) is false

CPT245

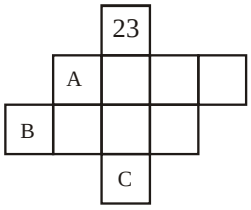
4. Total number of acidic oxides among N_2O_3, NO_2, N_2O, Cl_2O_7, SO_2, CO, CaO, Na_2O and NO is.

(1) 6 (2) 3

(3) 5 (4) 4

CPT246

5. Which of the following statement is not correct?



(1) Atomic no. of A, B, C are 40,57, 105 resp.

(2) Group no of A, B, C are IV B, III B and V B respectively

(3) Period no of A, B, C are, 4th, 5th and 6th resp.

(4) C is a radioactive element

CPT247

6. In elements, the incoming electron occupies which subshell after (n–1)d subshell :-

(1) np (2) ns

(3) (n–1)d (4) (n+1)s

CPT248

7. Which two elements are in same period as well as in same group of periodic table ?

(1) Z = 23, Z = 31

(2) Z = 65, Z = 66

(3) Z = 52, Z = 87

(4) Z = 58, Z = 46

CPT249

8. Which of the following statement is incorrect?

(1) Metallic radius of Mn is higher than that of Cr and Fe

(2) $I.E_I$ of O is greater than N

(3) Fe, Co, Ni have almost same atomic radius

(4) Irregular trend of atomic radius on moving from Sc to Zn

CPT250

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9. Which of the following statement is not correct

(1) Inert gases have a different number of outermost electrons than their group number

(2) In the sixth period the orbitals being filled are 6s, 4f, 5d, 6p

(3) The second transition series contains the elements from Y to Cd.

(4) O^{-2} has the same no. of e$^-$s in its outermost as well as penultimate shell

CPT251

10. Which of the following statement is not correct?

(1) Tc and Pm are not found in nature

(2) Element having $ns^2\ np^5$ configuration are placed in the column before, the extreme right in the P.T.

(3) Most of the man made element occurs in the actinoid series

(4) The 3d - transition series contains elements having atomic number from 21 to 29

CPT252

11. Which of the following statement is not correct?

(1) In the transition elements the incoming e$^-$ occupy (n–1)d subshell in preference to np.

(2) Elements having atomic number 57 to 71 belong to same group

(3) Lanthanum is the first element of Lanthanoid series.

(4) Actinium violates the Aufbau's principle

CPT253

12. Total number of elements of 6th period which have one or more than one 4d electrons :-

(1) 10 (2) 16

(3) 32 (4) Zero

CPT254

13. Suppose an orbital may accomodate 3 electrons then the number of elements in IV period :-

(1) 18 (2) 27

(3) 12 (4) 45

CPT255

14. Which of the following is not correctly matched?

(1) [Xe] $4f^{14}\ 5d^{10}\ 6s^2 \rightarrow$ Transition element

(2) [Rn] $5f^{14}\ 6d^1\ 7s^2 \rightarrow$ Inner transition element

(3) [Xe] $4f^{14}\ 5d^{10}\ 6s^2\ 6p^6\ 7s^2 \rightarrow$ Representative element

(4) [Xe] $4f^{14}\ 5d^2\ 6s^2 \rightarrow$ d-block element

CPT256

15. An element whose IUPAC name is ununtrium (Uut) belongs to :-

(1) s-block (2) p-block

(3) d-block (4) f-block

CPT257

16. Which of the following fact is not explainable by lanthanoid contraction ?

(1) I.E. of Ag < Au (2) $I.E._1$ of Hg > Cd

(3) Size of Hf ≃ Zr (4) Size of Y < La

CPT258

17. Which of the following property gradually increases from C to F ?

(1) I.P

(2) E.A.

(3) Shielding Constant

(4) All of these

CPT259

18. Which of the following statement is not correct?

(1) The first ionisation energies (in kJ mol^{-1}) of carbon, silicon, germanium, tin and lead are 1086, 786, 761, 708 and 715 respectively.

(2) Down the group, electronegativity decreases regularly from B to Tl in boron family

(3) Among oxides of the carbon family, CO is neutral, GeO_2 is acidic and SnO is amphoteric.

(4) The 4f-and 5f-inner transition elements are placed separately at the bottom of the periodic table to maintain its structure.

CPT260

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19. Select the correct statements of the following:-
(a) Effective nuclear charge for nitrogen is 3.90
(b) IP of Ne is more than Na^+
(c) Order of electronegativity- $sp > sp^2 > sp^3$
(d) Order of acidic character- $NH_3 < PH_3 < AsH_3$
(1) a, b, d
(2) b, c
(3) a, c, d
(4) a, b, c, d

CPT261

20. Which of the following statement is not correct ?
(1) EN of nitrogen on pauling scale is 3.0 in all nitrogen compounds
(2) The first I.E. for two isotopes of the same element has to be same
(3) Removal of electron from orbitals bearing higher n value is easier. than from orbital having lower n value.
(4) The size of isoelectric species is affected by nuclear charge (z)

CPT262

21. Which statement is/are incorrect ?
(1) In alkali metals in a group, from top to bottom increase in size is maximum from Na to K
(2) Addition of e^- in P atom will be exothermic
(3) IP of F is greater than its EA value
(4) Reaction $O^-_{(g)} + S_{(g)} \rightarrow O_{(g)} + S^-_{(g)}$ is endothermic

CPT263

22. According to Slater's rule, the set of elements that shows incorrect order of Z_{eff} are :-
(1) Al > Mg (2) Na < Al
(3) K > Na (4) Sc < Ti

CPT264

23. Lanthanoid contraction will not have any influence when we compare size of :-
(1) Zr & Hf (2) Y & La
(3) Mo & W (4) Tc & Re

CPT265

24. In which one of the following pairs the radius of the second specie is greater than that of the first ?
(1) O^{-2}, N^{-3} (2) Na, Mg
(3) Al, Be (4) Li^+, Be^{+2}

CPT266

25. Pair of smallest and largest species of radii among Mg, Al, Mg^{+2}, Al^{+3} is :-
(1) Mg^{+2}, Mg (2) Al^{+3}, Mg
(3) Mg^{+2}, Al (4) Al^{+3}, Al

CPT267

26. The order of basic character of given oxides is:-
(1) $Na_2O > MgO > Al_2O_3 > Cl_2O_7$
(2) $MgO > Al_2O_3 > Cl_2O_7 > Na_2O$
(3) $Al_2O_3 > MgO > Cl_2O_7 > Na_2O$
(4) $Cl_2O_7 > Na_2O > MgO > Al_2O_3$

CPT268

27. Correct order of Ionic radii :-
(1) $Ti^{4+} < Mn^{7+}$ (2) $Cr^{+6} > Cr^{3+}$
(3) $K^+ > Cl^-$ (4) $P^{3+} > P^{+5}$

CPT269

28. Which two are closest to one another in size ?
(1) Li^+ and Na^+
(2) Be^{+2} and Mg^{+2}
(3) Be^{+2} and Li^+
(4) Li^+ and Mg^{+2}

CPT270

29. Total number of elements of 5th period which have one or more than one 5d electrons :-
(1) 10 (2) 16
(3) 30 (4) Zero

CPT271

30. In which pair the first atom or ion is not larger than the second :-
(1) N, F (2) Cl^-, Cl
(3) O, S (4) Fe^{2+}, Fe^{3+}

CPT272

31. Incorrect order of radius is :-
(1) $Sr^{+2} < Rb^+ < Br^- < Se^{-2}$
(2) Ni > Cu > Zn
(3) $Co > Co^{+2} > Co^{+3} > Co^{+4}$
(4) $Ba^{+2} < Cs^+ < Se^{-2} < As^{-3}$

CPT273

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32. Which one of the following statements is incorrect ?
(1) Greater the nuclear charge, greater is the electron gain enthalpy
(2) Nitrogen has almost zero electron affinity
(3) Electron gain enthalpy decreases from fluorine to iodine in the group
(4) Chlorine has highest electron gain enthalpy
CPT274

33. Which of the following properties among halogens decrease(s) from fluorine to iodine ?
(1) Electronegativity
(2) Bond energy
(3) Ionisation energy
(4) (1) & (3) both
CPT275

34. Which of the following statements is incorrect ?
(1) The second ionization energy of sulphur is greater than that of chlorine
(2) The second ionization energy of magnesium is greater than that of sodium.
(3) The first ionization energy of aluminium is less than as that of gallium
(4) The second ionization energy for boron is greater than that of carbon
CPT276

35. Correct order of first ionisation energy is:-
(1) C > N > O (2) Mg > Al > P
(3) Li < Be < N (4) Tℓ > B > In
CPT277

36. From the ground state, electronic configuration of the elements given below, pick up the one with highest value of second ionization energy :-
(1) $1s^2\ 2s^2\ 2p^6\ 3s^2$ (2) $1s^2\ 2s^2\ 2p^6\ 3s^1$
(3) $1s^2\ 2s^2\ 2p^6$ (4) $1s^2\ 2s^2\ 2p^5$
CPT278

37. An element has successive ionization enthalpies as 940 (first), 2080, 3090, 4140, 7030, 7870, 16000 and 19500 kJ mol^{-1}. To which group of the periodic table does this element belong ?
(1) 14 (2) 15 (3) 16 (4) 17
CPT279

38. Read the following statements:
(I) Atomic size of following pair are almost same
Fe, Co; Al, Ga ; Zr, Hf
(II) E.A. order of Li, Be, B, C, N, O is
Be < N < B < Li < C < O
(III) Ionisation energy of
Al < Ga & Zr < Hf
(IV) Ionic radius
$O^{-2} < F^- < Na^+ < Mg^{+2}$
Select the correct statements(s)
(1) I, III, IV (2) I, II, III
(3) II, III, IV (4) All of the above
CPT280

39. The first ionisation energy of first atom is greater than that of second atom, whereas reverse order is true for their second ionisation energy. Which set of elements is not in accordance to above statement ?
(1) C, B (2) P, S
(3) B, Be (4) Mg, Na
CPT281

40. Correct order by electron affinity for the following element is :-
(1) F > Cl > O > S
(2) Cl > F > S > O
(3) Cl > F > O > S
(4) F > Cl > S > O
CPT282

41. Which of the following Incorrect statements(s) among the following is/are ?
(1) The first ionisation energy of calcium is more than first ionisation energy of Gallium
(2) The second ionisation energy of copper is greater than that of potassium
(3) The third ionisation energy of Mg is greater than the third ionisation energy of Al
(4) The IE_1 of Mg^+ is less than the IE_1 of Na^+
CPT283

42. Find the incorrect second ionisation energy order from following option :-
(1) Al > Mg (2) Te > Sb
(3) Fe > Fe^+ (4) In > Sr
CPT284

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43. If atomic number of an inert gas is Z then an element with which of the following atomic number will has highest I.E.

(1) Z – 2 (2) Z – 1
(3) Z + 1 (4) Z + 2

CPT285

44. If I.E. of Na, Mg and Si are respectively 496, 737 and 786 kJ mol^{-} then I.E. of Aℓ in KJ mol^{-} would be :-

(1) 575 (2) 760
(3) 390 (4) 1120

CPT286

45. Correct order of metallic character is :-

(1) P < Si < Be < Mg < Na
(2) Si < P < Be < Na < Mg
(3) P < Si < Mg < Be < Na
(4) Si < P < Mg < Na < Be

CPT287

46. Which one of the following statements is incorrect in relation to ionization enthalpy?

(1) Ionization enthalpy increases for the removing each successive electron.
(2) The greatest increase in ionization enthalpy is experienced on removal of electron from core noble gas configuration.
(3) End of valence electrons is marked by a big jump in ionization enthalpy.
(4) Removal of electron from orbitals bearing lower n value is easier than from orbital having higher n value.

CPT288

47. IP_1, IP_2, IP_3, IP_4 and IP_5 for an element is 7.1, 14.3, 34.5, 46.8 and 162.2 eV respectively then element is :-

(1) Na (2) Si (3) F (4) Ca

CPT289

48. Which process requires maximum energy ?

(1) $Na(g) \rightarrow Na^{+}(g) + e^{-}$
(2) $Al^{+3}(g) \rightarrow Al^{+4}(g) + e^{-}$
(3) $Al^{+2}(g) \rightarrow Al^{+3}(g) + e^{-}$
(4) $Na^{+}(g) \rightarrow Na^{+2}(g) + e^{-}$

CPT290

49. Amongst the following the incorrect order is:-

(1) $IE_1(Al) < IE_1(Mg)$
(2) $IE_1(Na) < IE_1(Mg)$
(3) $IE_2(Mg) > IE_2(Na)$
(4) $IE_1(Mg) > IE_1(Ca)$

CPT291

50. Which of the following statement is correct?

(1) Number of e^{-} in $(n-1)^{th}$ shell of f-block element are (0–1)
(2) 6th period contains 10 elements which have one or more 6d electrons
(3) In the 8th period, subshell used will be 8s, 5g, 6f, 7d, 8p
(4) Both (1) and (3)

CPT292

51. The increasing order of electron affinity of the electronic configurations of element is:-

(I) $1s^2 2s^2 2p^6 3s^2 3p^5$ (II) $1s^2 2s^2 2p^3$
(III) $1s^2 2s^2 2p^5$ (IV) $1s^2 2s^2 2p^6 3s^1$

(1) II < IV < III < I
(2) I < II < III < IV
(3) I < III < II < IV
(4) IV < III < II < I

CPT293

52. The element having very high ionization enthalpy but zero electron affinity is :-

(1) H (2) F (3) He (4) Be

CPT294

53. Which of the following statements is/are correct ?

(1) Van der Waals' radius of iodine is more than its covalent radius.
(2) All isoelectronic ions of corresponding elements belong to the same period of the periodic table
(3) The electron affinity of fluorine is greater than that of chlorine.
(4) IE_2 of N-atom is higher than that of O-atom

CPT295

54. Highest electron affinity observe in :-

(1) $2s^2 2p^5$ (2) $2s^2 2p^4$
(3) $2s^2 2p^3$ (4) $2s^2 2p^1$

CPT296

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55. Electron affinity of :
(1) carbon is greater than oxygen
(2) sulphur is lesser than oxygen
(3) iodine is higher than bromine
(4) bromine is lesser than chlorine
CPT297

56. Which of the following EA order is not correct
(1) N < P < O < S (2) Cl > O > N > C
(3) Ne < P < Si < S (4) B < C < Si < S
CPT298

57. I.P is maximum for :-
(1) Li^+ (2) Ne (3) Be (4) B
CPT299

58. Select the incorrect statements :
(1) Size of I^- is larger than F^-
(2) Rb is more electropositive compared to Ca
(3) Na^+ is more electronegative than the Na
(4) Cl^- is more electronegative than that of F
CPT300

59. Which is/are correct about electronegativity order of the following elements ?
(1) P > Si (2) C > N
(3) C > Br (4) Sr > Ca
CPT301

60. Select the incorrect statement(s) :
(1) In general more the ionisation energy more will be electronegativity
(2) Electronegativity increases means metallic character increases
(3) In general lower will be the ionisation energy, higher will be reducing property
(4) Cl has higher electron affinity than F
CPT302

61. An atom with high electronegativity generally has:-
(1) tendency to form +ve ions
(2) high ionisation potential
(3) large atomic size
(4) low electron affinity
CPT303

62. Which order is correct ?
(1) N < O (IP_2) (2) O > N (IP_1)
(3) Zn < Mn (IP_1) (4) Al > B (IP_1)
CPT304

63. p-block elements of 6th period are represented as :-
(1) $[Xe]\ 4f^{14}\ 5d^{10}\ 6p^{1-6}$
(2) $[Xe]\ 5f^{14}\ 6d^{10}\ 6p^{1-6}$
(3) $[Kr]\ 5f^{14}\ 5d^{10}\ 6p^{1-6}$
(4) $[Xe]\ 4f^{0}\ 5d^{10}\ 6p^{1-6}$
CPT305

64. Correct atomic size is :-
(1) Li > I > F < Cl < Br
(2) I > Li > F < Cl > Br
(3) I > Br > Cl > F > Li
(4) Li > F < I < Br > Cl
CPT306

65. Which of the following valence electron experiences maximum effective nuclear charge?
(1) $4s^1$ (2) $4p^1$ (3) $3d^1$ (4) $2p^3$
CPT307

66. Incorrect order of IP is :-
(1) $Cu^+ > Zn^+$ (2) Cu < Zn
(3) Tl < Al (4) Pd > Ag
CPT308

67. Lowest I^{st} IP is of :-
(1) Pb (2) Si (3) Sn (4) C
CPT309

68. Which of following radius order is not correct?
(1) Ti < Zr ≈ Hf
(2) Sc < Y ≈ La
(3) B < Al ≈ Ga < In ≈ Tl
(4) O < N < S < P
CPT310

69. Correct order of EA is :-
(1) Cl > O > N > C (2) Cl > O > C > N
(3) Cl > N > C > O (4) Cl > C > O > N
CPT311

70. To which of the following atom, the attachment of electron is most difficult ?
(1) Radon (2) Nitrogen
(3) Oxygen (4) Radium
CPT312

71. Select process having exothermic step:
(1) $S^-_{(g)} \longrightarrow S^{2-}_{(g)}$ (2) $O_{(g)} \longrightarrow O^-_{(g)}$
(3) $N_{(g)} \longrightarrow N^-_{(g)}$ (4) $Al^{2+}_{(g)} \longrightarrow Al^{3+}_{(g)}$
CPT313

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	**4**	**2**	**1**	**4**	**3**	**1**	**2**	**2**	**4**	**4**	**3**	**3**	**2**	**1**	**2**
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	**4**	**3**	**2**	**3**	**1**	**4**	**3**	**2**	**1**	**2**	**1**	**4**	**4**	**4**	**3**
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	**2**	**3**	**4**	**2**	**3**	**2**	**3**	**2**	**3**	**2**	**2**	**3**	**2**	**1**	**1**
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	**4**	**2**	**2**	**3**	**3**	**1**	**3**	**1**	**1**	**4**	**2**	**1**	**4**	**1**	**2**
Question	61	62	63	64	65	66	67	68	69	70	71				
Answer	**2**	**1**	**1**	**1**	**4**	**3**	**3**	**2**	**2**	**1**	**2**				

Chemical Bonding and Molecular Structure – Chemical Bonding

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1. Order of Covalent bond;
A. KF > KI; LiF > KF
B. KF < KI; LiF > KF
C. $SnCl_4 > SnCl_2$; CuCl > NaCl
D. LiF > KF; CuCl < NaCl
E. KF < KI; CuCl > NaCl
(1) C, E only (2) B, C only
(3) B, C, E only (4) A, B only
CCB371

2. What is the number of unpaired electron(s) in the highest occupied molecular orbital of the following species : N_2 : N_2^+; O_2; O_2^+ ?
(1) 0, 1, 2, 1 (2) 2, 1, 2, 1
(3) 0, 1, 0, 1 (4) 2, 1, 0, 1
CCB372

3. The total number of lone pairs of electrons on oxygen atoms of ozone is
(1) 6 (2) 4
(3) 5 (4) 2
CCB373

4. The alkaline earth metal sulphate(s) which are readily soluble in water is/are:
(A) $BeSO_4$ (B) $MgSO_4$
(C) $CaSO_4$ (D) $SrSO_4$
(E) $BaSO_4$
Choose the correct answer from the options given below:
(1) A only (2) B only
(3) A and B (4) B and C
CCB374

5. Amongst the following, the number of species having the linear shape is .
$XeF_2, I_3^+, C_3O_2, I_3^-, CO_2, SO_2, BeCl_2$ and $BCl_2^{\ominus}$
(1) 4 (2) 6
(3) 2 (4) 5
CCB375

6. According to MO theory the bond orders for O_2^{2-}, CO and NO^+ respectively, are
(1) 1, 3 and 3 (2) 1, 3 and 2
(3) 1, 2 and 3 (4) 2, 3 and 3
CCB376

7. The correct order of bond enthalpy ($kJ\ mol^{-1}$) is :
(1) Si - Si > C - C > Sn - Sn > Ge - Ge
(2) Si - Si > C - C > Ge - Ge > Sn - Sn
(3) C - C > Si - Si > Sn - Sn > Ge – Ge
(4) C - C > Si - Si > Ge - Ge > Sn - Sn
CCB377

8. Two hybrid orbitals have a bond angle of 120°. The percentage of p character in the hybrid orbital is nearly:-
(1) 25% (2) 33% (3) 50% (4) 66%
CCB378

9. A molecule XY_2 contains two σ, two π bond and one lone pair of electron in the valence shell of X. The arrangement of lone pair as well as bond pair is :
(1) Square pyramidal
(2) Linear
(3) Trigonal planar
(4) Unpredictable
CCB379

10. In CO_2, SO_2, SiO_2 each central atom is covalently bonded with :-
(1) 2, 2, 4 oxygen atoms respectively
(2) 2, 2, 2 oxygen atoms respectively
(3) 2, 4, 4 oxygen atoms respectively
(4) 4, 4, 4 oxygen atoms respectively
CCB380

11. Which of the following statement is correct, if % s-character of a hybrid orbital decreases
(1) The bond angle decreases
(2) The bond strength increases
(3) The bond length decreases
(4) Size of orbital decreases
CCB381

12. Correct order of bond angle is :-
(1) $CH_3^+ > CH_3^- > CH_4$ (2) $BeCl_2 > BF_3 < CF_4$
(3) $NH_4^+ > NH_3 > NH_2^-$ (4) $OF_2 > OCl_2 > OBr_2$
CCB382

13. In which pair both are not isostructural ?
(1) BH_4^- & AlH_4^- (2) NH_4^+ & PH_4^+
(3) PCl_6^- & SiF_6^{-2} (4) BCl_4^- & ICl_4^-
CCB383

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14. Select in which both have sea-saw shape ?
(1) XeO_2F_2, SiF_4 (2) XeO_2F_2, $IO_2F_2^-$
(3) $TeCl_4$, ICl_4^- (4) BrO_3F, $XeOF_2$
CCB384

15. Which of the following molecule is planar as well as polar ?
(1) PCl_3 (2) SF_4
(3) ClF_3 (4) XeO_3F_2
CCB385

16. Which of the following specie is non polar with presence of polar bond and lone pair of electron on central atom.
(1) CO_2 (2) SF_4 (3) XeF_4 (4) CF_4
CCB386

17. Total no. of vacant orbitals in valence shell of sulphur when it undergoes formation of SF_4:-
(1) 5 (2) 4 (3) 3 (4) 2
CCB387

18. Correct order of extent strength of overlapping is :-
(1) 2s – 2s > 2p – 2p > 2s – 2p
(2) 1s – 1s > 2p – 2p > 2s – 2p
(3) 1s – 1s < 2s – 2s < 3s – 3s
(4) 3s – 3s > 3s – 3p > 3p – 3p
CCB388

19. Dative Bond is present in :-
(1) KI_3 (2) KNO_2
(3) KHF_2 (4) All
CCB389

20. Which of the following element never form compound in ground state ?
(1) N (2) B (3) C (4) P
CCB390

21. Which of the following pair have same hybridisation ?
(1) ClO_4^- & ClO_2^- (2) SF_4 & CCl_4
(3) BF_3 & NF_3 (4) CO_2 & SO_2
CCB391

22. Which orbital is not involved in the formation of PCl_5 molecule ?
(1) s (2) d_{z^2} (3) $d_{x^2-y^2}$ (4) p_z
CCB392

23. Shape of molecule having 4-bond pair and one lone pair is :-
(1) Trigonal bipyramidal
(2) T-Shape
(3) See-saw
(4) Square planar
CCB393

24. Which of the following molecule has both pπ-pπ and pπ-dπ bonds
(1) XeO_2F_2 (2) SO_2Cl_2
(3) SO_3 (4) XeO_3
CCB394

25. Intramolecular hydrogen bonding is not present in :-
(1) p-fluoro phenol
(2) Chloral hydrate
(3) o-nitro phenol
(4) o-fluoro phenol
CCB395

26. Among the following Van der waals force is maximum in :-
(1) HBr (2) LiBr (3) LiCl (4) AgBr
CCB396

27. The correct expected order of decreasing lattice energy is :-
(1) CaO > $MgBr_2$ > CsI
(2) $MgBr_2$ > CaO > CsI
(3) CsI > $MgBr_2$ > CaO
(4) CsI > CaO > $MgBr_2$
CCB397

28. In which of the following set do all the three compounds have bonds that are mainly ionic?
(1) NaCl, NCl_3, CCl_4
(2) CsBr, $BaBr_2$, SrO
(3) CsF, BF_3, NH_3
(4) Al_2O_3, CaO, SO_2
CCB398

29. Which of the following order is not correct?
(1) CrO < Cr_2O_3 < CrO_3 (covalent character)
(2) ZnO > ZnS (ionic character)
(3) $BeCl_2$ > $MgCl_2$ > $CaCl_2$ > $SrCl_2$ > $BaCl_2$ (ionic character)
(4) KCl < AgCl (covalent character)
CCB399

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30. Which of the following order is incorrect for thermal stability ?
(1) $BeO > MgO > CaO$
(2) $NH_3 < H_2O < HF$
(3) $HF < HCl < HBr < HI$
(4) $Li_2SO_4 < Na_2SO_4 < K_2SO_4 < Rb_2SO_4 < Cs_2SO_4$

CCB400

31. Which of the following solubility order is not correct ?
(1) $MgF_2 < CaF_2 < SrF_2 < BaF_2$
(2) $Be(OH)_2 < Mg(OH)_2 < Ca(OH)_2 < Sr(OH)_2$
(3) $Li_2O < Na_2O < K_2O < Rb_2O < Cs_2O$
(4) $CsF < CsCl < CsBr < CsI$

CCB401

32. Which show highest lattice energy ?
(1) Al_2O_3 (2) TiC
(3) CaF_2 (4) NaF

CCB402

33. Which of the following will be most covalent?
(1) $NaCl$ (2) Na_2S
(3) $MgCl_2$ (4) MgS

CCB403

34. Ionic mobility of $Li^+_{(aq)}$ is less than $Na^+_{(aq)}$ and $K^+_{(aq)}$ because
(1) ionisation potential of lithium is low
(2) charge density of Li^+ is low
(3) high hydration tendency of Li^+
(4) Li^+ keeps two electrons

CCB404

35. Which of the metal carbonates is thermally most stable ?
(1) $MgCO_3$ (2) $BaCO_3$
(3) $SrCO_3$ (4) K_2CO_3

CCB405

36. Which of the following halides has the maximum covalent character?
(1) LiI (2) NaI (3) KI (4) CsI

CCB406

37. Magnesium oxide can be obtained by heating :-
(1) $MgCO_3$ (2) $Mg(NO_3)_2$
(3) $Mg(OH)_2$ (4) All

CCB407

38. The incorrect order of bond angle :-
(1) $CO_2 > CO_3^{2-} > CF_2Cl_2$
(2) $NO_2^+ > NO_3^- > NO_2^-$
(3) $XeF_2 > XeO_3 > XeO_4$
(4) $PH_3 > AsH_3 > SbH_3$

CCB408

39. Which of the following molecule has two lone pairs and bond angle < 109°28' ?
(1) SF_2 (2) KrF_4
(3) ICl_4^- (4) All of these

CCB409

40. In which of the following compound all the bond angles are equal :-
(1) SF_4 (2) CCl_4 (3) $CHCl_3$ (4) XeF_6

CCB410

41. Which of the following do not exhibit resonance ?
(1) CO_3^{-2} (2) ClO_3^- (3) SCl_2 (4) SO_3^{-2}

CCB411

42. Which of the following is not correctly matched between given species and type of overlapping ?
(1) XeO_3 : Three ($d\pi-p\pi$) bonds
(2) H_2SO_4: Two ($d\pi-p\pi$) bonds
(3) SO_2 : Two ($d\pi-p\pi$) bonds
(4) $HClO_4$: Three ($d\pi-p\pi$) bonds

CCB412

43. Which of the following pair of species are iso structural ?
(1) ICl_2^- & I_3^+ (2) NO_2 & CO_2
(3) XeF_4, ICl_4^- (4) SO_3^{-2}, ClO_4^-

CCB413

44. Which of the following pairs of species have largest difference in spin only magnetic moment ?
(1) O_2, O_2^+ (2) O_2, O_2^{2-}
(3) O_2^+, O_2^{2-} (4) O_2^-, O_2^+

CCB414

45. In $ICl_4^{\ominus}$, shape is square planar. The number of bond pair – lone pair repulsion at 90° are:-
(1) 6 (2) 4 (3) 12 (4) 8

CCB415

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46. Which of the following set of molecules have same shape but different hybridisation ?
(1) XeO_4, SF_4 (2) XeO_3, BrF_3
(3) H_2O, $SnCl_2$ (4) I_3^-, I_3^+

CCB416

47. Which of the following order of Bond strength is not correct ?
(1) Diamond < Graphite (C–C bond)
(2) $SO_2 > SO_3^{-2} > SO_4^{-2}$ (S–O bond)
(3) $CO_3^{-2} < HCO_2^-$ (C–O bond)
(4) $O_2^- < O_2 < O_2^+$ (O–O bond)

CCB417

48. The incorrect order of bond angle :-
(1) $CO_2 > CO_3^{2-} > CF_2Cl_2$
(2) $NO_2^+ > NO_3^- > NO_2^-$
(3) $XeF_2 > XeO_3 > XeO_4$
(4) $PH_3 > AsH_3 > SbH_3$

CCB418

49. Which of the following does not contain PX_4^+ type cation in solid phase ?
(X=halogen atom)
(1) PF_5 (2) PCl_5
(3) PBr_5 (4) None of these

CCB419

50. Correct order of Bond angle is
(1) $NO_2^+ < NO_2 < NO_2^-$ (2) $BeCl_2 > BF_3 < CF_4$
(3) $NH_4^+ > NH_3 > NH_2^-$ (4) $OF_2 > OCl_2 > OBr_2$

CCB420

51. Which of the following statements is incorrect?
(1) $C\ell F_3$ has T-shape
(2) In SF_4, F-S-F equatorial bond angle is less than 120°
(3) In $[IC\ell_4]^-$, $C\ell$-I-$C\ell$ bond angle is 109°
(4) Shape of XeF_2 molecule is linear

CCB421

52. Identify the correct option :-
(1) $NO_3^- > NH_3 > NH_2^-$ (order of bond angle)
(2) $(CH_3)_3$ B is a planar molecule
(3) In NH_4Cl, 'N' atom is in sp^3d Hybridisation
(4) $BF_3 < BCl_3 < BBr_3 < BI_3$ (Order of bond angle)

CCB422

53. In which of the following molecules no. of lone pairs and bond pairs on central atom are not equal ?
(1) H_2O (2) I_3^- (3) O_2F_2 (4) SCl_2

CCB423

54. Number of carbon atom present linearly in C_3O_2 :-
(1) 2 (2) 3 (3) 6 (4) 1

CCB424

55. In which of the following molecule all bond length are equal ?
(1) SF_4 (2) B_2H_6
(3) PCl_5 (4) SiF_4

CCB425

56. Which of the following has different hybridization than other ?
(1) I_3^- (2) XeO_3 (3) $SeCl_4$ (4) AsF_4^-

CCB426

57. Which of the following molecule exhibit $P_\pi - P_\pi$ bonding ?
(1) CO_3^{-2} (2) NO_3^{-1}
(3) SiO_2 (4) (1) and (2) both

CCB427

58. Which of the following statement is not correct?
(1) PCl_3F_2 has zero dipole moment
(2) PH^+_4 is having tetrahedral geometry with sp^3 hybridisation of central atom.
(3) All diatomic molecules with polar bonds have dipole moment.
(4) Four half filled pure orbital of carbon form same kind of bonds in CH_4

CCB428

59. Which of the following have same geometry?
(1) SbF_5^{2-}, XeF_5^-, IF_5
(2) SF_4, $XeOF_4$, PCl_4^+
(3) SF_6, PF_6^-, IOF_5
(4) $COCl_2$, $SOCl_2$, $XeOF_2$

CCB429

60. Out of $CHCl_3$, CH_4 and SF_4, the molecule are not having regular geometry are :-
(1) $CHCl_3$ only
(2) $CHCl_3$ & SF_4
(3) CH_4 only
(4) CH_4 & SF_4

CCB430

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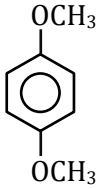
61. Which of the following will have pyramidal shape ?
(1) XeO_3 (2) ICl_4^-
(3) ICl_2^- (4) All of these
CCB431

62. Which of the following is electron deficient compound ?
(1) PF_3 (2) ClF_3
(3) AlF_3 (4) BF_3
CCB432

63. Maximum number of identical bond lengths are present in :-
(1) SF_6 (2) IF_7
(3) PCl_5 (4) SO_4^{-2}
CCB433

64. Which of the following compounds give paramagnetic gas on decomposition ?
(i) $Pb(NO_3)_2$ (ii) $LiNO_3$
(iii) $NaNO_3$ (iv) NH_4NO_2
(1) I, II, III (2) II, III
(3) I, II (4) III, IV
CCB434

65. Arrange the following compound in order of increasing dipole moment :-
(I) 1, 3, 5-Trichloro benzene
(II) 1, 2, 4-Trichloro benzene
(III) 1, 2, 3, 4-Tetrachloro benzene
(IV) P-dichloro benzene
(1) I = IV < II < III (2) IV < I < II < III
(3) IV = I < III < II (4) IV < II < I < III
CCB435

66. Which of the following molecule has some dipole moment ?
(1) 
(2) 
(3) PF_3Cl_2 (4) All
CCB436

67. Which of the following species are Hypovalent ?
(I) $C\ell O_4^-$ (II) BF_3 (III) SO_4^{-2} (IV) CO_3^{-2}
(1) I, II, III (2) Only II
(3) II, IV (4) I, III, IV
CCB437

68. Which have maximum number of lone pair on central atom ?
(1) SO_2 (2) SCl_2
(3) XeF_2 (4) XeO_2F_2
CCB438

69. Which of the following statement is incorrect:-
(1) Both CO_2 and SiO_2 are linear
(2) If internuclear axis is Z-axis then p_z orbital overlaps to form σ bond
(3) The average formal charge on each 'O' atom in ClO_4^- is $-1/4$.
(4) $3d_{xy}-2p_x$ colateral overlapping is stronger than $3p_x$-$3p_x$ sideway overlapping
CCB439

70. Which of the following covalency of Xe is possible :-
(a) 2 (b) 4 (c) 5 (d) 8
(1) a, b only (2) a, b, d
(3) c only (4) b, d
CCB440

71. Select the correct statement :
(1) The order of (C–O) B.O. in various species $CO_3^{2-} < CO_2 < CO$
(2) PH_5 can undergo sp^3d hybridisation to have octahedral geometry
(3) Dipole moment of CH_3F is greater than of CH_3Cl
(4) Hydrogen bonding in N—H---N is more as compared to F—H---F
CCB441

72. Intramolecular hydrogen bonding is not present in :-
(1) Urea (2) Chloral hydrate
(3) Salicyladehyde (4) o-fluoro phenol
CCB442

73. Which of the following order of H-bond strength is not correct ?
(1) $HF > H_2O > NH_3 > H_2S$
(2) $D_2O > H_2O$
(3) $H_2O > H_2O_2$
(4) $HF_2^- < (HF)_2$
CCB443

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74. Dipole-dipole (Keesom) attraction is present in :-

(1) $KCl + H_2O$

(2) $CH_3—\underset{\underset{O}{\|}}{C}—CH_3 + CH_3—C\equiv N$

(3) $NO_3^- + Cl_2$

(4) $HCl + Cl_2$

CCB444

75. The solubility of inert gases in water is due to:-

(1) Keesom attraction

(2) Debye attraction

(3) London force

(4) Ion-dipole attraction

CCB445

76. Stability of the species Li_2, Li_2^- and Li_2^+ increases in the order of :-

(1) $Li_2^- < Li_2^+ < Li_2$

(2) $Li_2 < Li_2^+ < Li_2^-$

(3) $Li_2 < Li_2^- < Li_2^+$

(4) $Li_2^- < Li_2 < Li_2^+$

CCB446

77. Which of the following compound does not exist?

(1) PCl_5 (2) OF_6

(3) OF_2 (4) BrF_5

CCB447

78. How Van der waals force is related to molecular weight :-

(1) Inversly

(2) Independent

(3) Directly proportional

(4) None of these

CCB448

79. Which of the following can not be calculated by Born Haber cycle in the formation of MgO?

(1) Lattice energy of MgO

(2) EA of oxygen

(3) Hydration energy of Mg^{+2}

(4) I.E. of Mg

CCB449

80. Which pair is not isomorphous pair ?

(1) Green vitriol, Epsum salt

(2) KNO_3, $CaCO_3$

(3) NaF, Na_2O

(4) Potash alum, chrome alum

CCB450

81. How many compounds show conductivity ?

HCl(aq.), $AlCl_3$ (molten), AlF_3 (molten), Graphite, Na(s), $AlCl_3$(aq.)

(1) 4 (2) 5 (3) 6 (4) 3

CCB451

82. Which of the following B.P. order is not correct?

(1) $SiF_4 < SiCl_4$ (2) $CH_4 < SiH_4 < GeH_4$

(3) $CH_4 < CD_4$ (4) He > Xe

CCB452

83. Which of the following order is not correct?

(1) $H_2O > NH_3$ (melting point)

(2) $CH_3–O–CH_3 < C_2H_5–OH$ (boiling point)

(3) $(CH_3)_2SO_4 > H_2SO_4$ (volatility)

(4) HCl > HF (Boiling point)

CCB453

84. In which of the following process, weaking of covalent bond take place :-

(1) Boiling of H_2O (2) Melting of KCN

(3) Boiling of CF_4 (4) Melting of SiO_2

CCB454

85. The species having diamagnetic nature and bond order 1.0 is :-

(1) O_2^{2-} (2) O_2^+ (3) O_2^{2+} (4) O_2

CCB455

86. Which specie doesn't exist ?

(1) B_2 (2) Be_2 (3) Li_2 (4) H_2^+

CCB456

87. Which is the correct decreasing order of covalent nature ?

(1) $SiCl_4 > AlCl_3 > ZnCl_2 > CaCl_2 > MgCl_2$

(2) $SiCl_4 > AlCl_3 > MgCl_2 > CaCl_2 > ZnCl_2$

(3) $ZnCl_2 > SiCl_4 > AlCl_3 > MgCl_2 > CaCl_2$

(4) $SiCl_4 > AlCl_3 > ZnCl_2 > MgCl_2 > CaCl_2$

CCB457

88. Which of following will not conduct electricity?

(1) KOH (aq.) (2) Fused NaCl

(3) Solid KCl (4) Graphite

CCB458

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89. Which of the following statement is not correct?
(1) Ortho nitro phenol is more volatile as compared to para nitro phenol.
(2) HF(s) has zig-zag structure
(3) HF is more volatile than HCl.
(4) Ethylene glycol is less viscous than glycerol.

CCB459

90. Which of the following condition is true for a compound to be soluble ?
(L.E.=Lattice Energy; H.E.=Hydration Energy)
(1) |L.E.| > |H.E.| (2) |L.E.| = |H.E.|
(3) |L.E.| < |H.E.| (4) None of these

CCB460

91. If Hunds rule is not applicable then which of the following specie is paramagnetic ?
(1) B_2 (2) N_2 (3) O_2^{-1} (4) O_2

CCB461

92. The salt having least solubility in water :-
(1) $CaSO_4$ (2) $BeSO_4$
(3) $MgSO_4$ (4) $BaSO_4$

CCB462

93. The fluoride which is most soluble in water:-
(1) CaF_2 (2) BaF_2 (3) SrF_2 (4) BeF_2

CCB463

94. Which of the following compound will release brown color gas on heating ?
(1) $CaCO_3$ (2) $NaNO_3$
(3) $Zn(NO_3)_2$ (4) $NaHCO_3$

CCB464

95. Correct solubility order :-
(1) $CaCrO_4 > BaCrO_4$
(2) $BeCO_3 < BaCO_3$
(3) $LiNO_3 < CsNO_3$
(4) $NaClO_4 < KClO_4$

CCB465

96. Which of the following halide can be converted into anhydrous form on heating?
(1) $BeCl_2 \cdot H_2O$ (2) $KCl \cdot 6H_2O$
(3) $AlCl_3 \cdot 6H_2O$ (4) $FeCl_3 \cdot 6H_2O$

CCB466

97. Which of the following thermal stability order is not correct ?
(1) $NaHCO_3 < KHCO_3 < RbHCO_3$
(2) $CaO_2 > SrO_2 > BaO_2$
(3) $Na_2SO_4 > MgSO_4 > Al_2(SO_4)_3$
(4) $CCl_4 > SiCl_4 > GeCl_4 > SnCl_4$

CCB467

98. Which of the following compound give metal and oxygen gas at high temp :-
(1) $NaNO_3$ (2) Ag_2CO_3
(3) K_2CO_3 (4) Li_2CO_3

CCB468

99. In which of the following species presence of lone pair does not affect idealized bond angle?
(I) PF_3 (II) BrF_3 (III) IF_5 (IV) $IC\ell_4^-$
(V) XeF_2
(1) I, II, III (2) IV, V
(3) II, V (4) None of these

CCB469

100. The incorrect order of bond angle :-
(1) $CO_2 > CO_3^{-2} > CF_2Cl_2$
(2) $NF_3 > NH_3 > NCl_3$
(3) $NO_2^+ > NO_3^- > NO_2^-$
(4) $PH_3 > AsH_3 > SbH_3$

CCB470

101. Which of the following is correct decreasing order of bond length ?
(1) P–O > S–O > Cl–O
(2) P–O > Cl–O > S–O
(3) S–O > Cl–O > P–O
(4) Cl–O > S–O > P–O

CCB471

102. Which of the following species are hypervalent?
(A) ClO_4^- (B) BF_3
(C) SO_4^{-2} (D) CO_3^{-2}
(1) A, B, C (2) C, D
(3) A, C (4) A, B

CCB472

103. The correct order of increasing C–O bond length of CO, CO_3^{-2} and CO_2 is :-
(1) $CO_3^{-2} < CO_2 < CO$
(2) $CO < CO_2 < CO_3^{-2}$
(3) $CO < CO_3^{-2} < CO_2$
(4) $CO_2 < CO_3^{-2} < CO$

CCB473

104. The specie having pyramidal shape is :-
(1) OSF_2 (2) BrF_3 (3) ClO_2^{-1} (4) SO_3

CCB474

105. PCl_5 exist but NCl_5 does not because :-
(1) Nitrogen atom is much smaller than P
(2) NCl_5 is unstable
(3) Nitrogen has no vacant d-orbitals
(4) Nitrogen is highly inert

CCB475

106. The correct order of increasing s-character (in percentage) in the hybrid orbitals of following molecules/ions is :-
(A) CO_3^{-2} (B) NCl_3 (C) $BeCl_2$
(1) B < A < C (2) A < B < C
(3) C < A < B (4) C < B < A

CCB476

107. Among $MgCl_2$, RbCl, $BeCl_2$ and LiCl. The compounds with highest and lowest ionic characters are :-
(1) RbCl and $BeCl_2$
(2) $MgCl_2$ and $BeCl_2$
(3) $BeCl_2$ and $MgCl_2$
(4) RbCl and LiCl

CCB477

108. Which of the following is correct order of dipole moment ?
(1) $NH_3 > NF_3$
(2) $BF_3 > BCl_3$
(3) $CF_4 > CH_4$
(4) $XeF_2 > XeO_3F_2$

CCB478

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	**3**	**1**	**1**	**3**	**4**	**1**	**4**	**4**	**3**	**1**	**1**	**3**	**4**	**2**	**3**
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	**3**	**4**	**2**	**1**	**2**	**1**	**3**	**3**	**3**	**1**	**4**	**1**	**2**	**3**	**3**
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	**4**	**2**	**4**	**3**	**4**	**1**	**4**	**3**	**4**	**2**	**3**	**3**	**3**	**2**	**4**
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	**3**	**2**	**3**	**1**	**3**	**3**	**1**	**2**	**2**	**4**	**2**	**4**	**4**	**3**	**2**
Question	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Answer	**1**	**4**	**1**	**1**	**1**	**4**	**2**	**3**	**1**	**2**	**1**	**1**	**4**	**2**	**2**
Question	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Answer	**1**	**2**	**3**	**3**	**3**	**2**	**4**	**4**	**4**	**1**	**2**	**4**	**3**	**3**	**3**
Question	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105
Answer	**3**	**4**	**4**	**3**	**1**	**2**	**2**	**2**	**2**	**2**	**1**	**3**	**2**	**1**	**3**
Question	106	107	108												
Answer	**1**	**1**	**1**												

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p-Block Elements - Group 13 to 18

1. Sum of π-bonds present in peroxodisulphuric acid and pyrosulphuric acid is
(1) 8 (2) 4 (3) 5 (4) 6

CPB298

2. Reaction of thionyl chloride with white phosphorus forms a compound [A], which on hydrolysis gives [B], a dibasic acid. [A] and [B] are respectively
(1) P_4O_6 and H_3PO_3 (2) PCl_3 and H_3PO_3
(3) PCl_5 and H_3PO_4 (4) $POCl_3$ and H_3PO_4

CPB299

3. An oxyacid among the following that contains both P-OH and P-P bonds simultaneously is :-
(1) $H_4P_2O_5$ (2) $H_4P_2O_7$
(3) $H_4P_2O_6$ (4) None of the above

CPB300

4. Bond dissociation energy of E-H bond of the "H_2E" hydrides of group 16 elements (given below), follows order.
(A) O (B) S (C) Se (D) Te
(1) A > B > C > D (2) A > B > D > C
(3) B > A > C > D (4) D > C > B > A

CPB301

5. The main factor responsible for weak acidic nature of B-F bond in BF_3 is :-
(1) Large electronegativity of F
(2) Three centred two electron bonds in BF_3
(3) pπ - dπ back bonding
(4) pπ–pπ back bonding

CPB302

6. Which of the following statement is true on the basis of back bonding ?
(1) Si-O bond is stronger than C–O bond
(2) Dimethyl ether acts as a better lewis base but not disilyl ether (SiH_3-O-SiH_3)
(3) $(CH_3)_3C$–O–H is less acidic than $(CH_3)_3Si$-OH
(4) All of these

CPB303

7. There is no S-S bond in :-
(1) $S_2O_3^{-2}$ (2) $S_2O_5^{-2}$
(3) $S_2O_6^{-2}$ (4) $S_2O_7^{-2}$

CPB304

8. Determine the correct order of stability :-
(1) $Al^{+1} > Al^{+3}$ (2) $Sn^{+2} > Sn^{+4}$
(3) $Pb^{+2} > Pb^{+4}$ (4) $Bi^{+3} < Bi^{+5}$

CPB305

9. Which of the following is not correctly matched for solid state of following compound?
(1) $AlCl_3$ - nonplanar, all bonds are not identical
(2) SO_3 - pπ-dπ bonding present, sp^3 hybridisation
(3) PBr_5 - sp^3 and sp^3d^2 hybridisation
(4) $BeCl_2$ - Complete octet, nonplanar

CPB306

10. Number of P–OH bonds is sodium dihydrogen pyrophosphate :-
(1) 4 (2) 3 (3) 2 (4) 0

CPB307

11. Which of the following pyroacid has X-X bond
(1) $H_2S_2O_7$ (2) $H_2S_2O_5$
(3) $H_4P_2O_5$ (4) $H_4P_2O_6$

CPB308

12. In which of the following options all species contain X–O–X bonds in structure (X = central atom) ?
(1) $H_2S_2O_5$, S_3O_9, $S_2O_6^{-2}$
(2) P_4O_{10}, P_4O_6, $H_3P_3O_9$
(3) N_2O_5, N_2O, N_2O_4
(4) $H_4P_2O_7$, $H_4P_2O_6$, $H_4P_2O_5$

CPB309

13. Order of stability of halogen oxides :-
(1) I < Cl < Br (2) I > Cl > Br
(3) I < Cl > Br (4) I > Cl < Br

CPB310

14. $(Si_2O_5)_n^{2n-}$ anion is obtained when :
(1) No oxygen of a SiO_4^{4-} tetrahedron is shared with another tetrahedron
(2) One oxygen of a SiO_4^{4-} tetrahedron is shared with another tetrahedron
(3) Two oxygen of a SiO_4^{4-} tetrahedron are shared with another SiO_4^{4-} tetrahedron
(4) Three oxygen of a SiO_4^{4-} tetrahedron are shared with another SiO_4^{4-} tetrahedron

CPB311

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15. BCl_3 does not exist as dimer but BH_3 exist as dimer (B_2H_6) because :-
(1) Chlorine is more electronegative than hydrogen
(2) There is pπ-pπ back bonding in BCl_3 but BH_3 does not contain such bonding
(3) Large sized chlorine atoms do not fit between the small boron atoms whereas small sized hydrogen atoms get fitted between boron atoms.
(4) None of these

CPB312

16. $Na_2S_2O_3$ is oxidized by I_2 to give :-
(1) Na_2S (2) $NaHSO_3$
(3) Na_2SO_4 (4) $Na_2S_4O_6$

CPB313

17. Product formed when PCl_5 react with conc. H_2SO_4 :-
(1) $SOCl_2$ (2) SO_2Cl_2 (3) S_2Cl_2 (4) SO_2

CPB314

18. P–O–P bond is absent in :-
(1) Dimetaphosphoric acid
(2) Pyro phosphorous acid
(3) Cyclic trimetaphosphoric acid
(4) Hypo phosphorous acid

CPB315

19. Which of the following is incorrect for H_3BO_3?
(1) It has a layer structure in which H_3BO_3 units are joined by hydrogen bonds.
(2) It is obtained by treating borax with conc. H_2SO_4.
(3) It does not act as a proton donor but acts as an acid by accepting hydroxyl ions.
(4) It can't acts as mild antiseptic.

CPB316

20. Maximum P-O-P linkage present in :-
(1) P_4O_6
(2) P_4O_{10}
(3) $(HPO_3)_3$
(4) (1) & (2) both have same

CPB317

21. Which of the following oxide does not give any oxyacid in aqueous solution ?
(1) N_2O_5 (2) Cl_2O_7
(3) CrO_3 (4) N_2O

CPB318

22. Number of peroxide linkage in CrO_5, H_2SO_5, $H_2S_2O_8$ are respectively :-
(1) 2, 1, 2 (2) 2, 2, 2
(3) 1, 1, 1 (4) 2, 1, 1

CPB319

23. Which of the following salt does not give any ppt with $AgNO_3$ sol.:
(1) F^- salt (2) Cl^- salt
(3) Br^- salt (4) I^- salt

CPB320

24. Find the incorrect match :-
(1) Al_2Cl_6 : 3c-4e bond is present
(2) $Al_2(CH_3)_6$: All carbon atoms are sp^3 hybridised
(3) I_2Cl_6 : Nonplanar
(4) B_2H_6 : Nonpolar

CPB321

25. Predominantly ionic halide of N-family is :-
(1) BiF_3 (2) $BiCl_5$ (3) PF_3 (4) NF_3

CPB322

26. In the following reaction
$2X + B_2H_6 \rightarrow [BH_2(X)_2]^+ [BH_4]^-$
then X will be :-
(1) NH_3 (2) CH_3NH_2
(3) $N(CH_3)_2H$ (4) All of these

CPB323

27. Which of the following reaction is not feasible?
(1) $F_2 + 2Cl^- \longrightarrow 2F^- + Cl_2$
(2) $Cl_2 + 2Br^- \longrightarrow 2Cl^- + Br_2$
(3) $Br_2 + 2F^- \longrightarrow 2Br^- + F_2$
(4) $Br_2 + 2I^- \longrightarrow 2Br^- + I_2$

CPB324

28. Which of the following order is not correct:-
(1) HI > HF > HBr > HCl melting point
(2) HF > HI > HBr > HCl boiling point
(3) HF < HCl < HBr < HI pK_a
(4) HF < HCl < HBr < HI reducing strength

CPB325

29. Which of the following is a set of reducing agents :-
(1) HNO_3, Fe^{+2}, Fe (2) $F^-, C\ell^-, MnO_4^-$
(3) I^-, Na, Fe^{+2} (4) $Cr_2O_7^{-2}$, NaCl

CPB326

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30. Which of the following compound is used for estimation of CO ?
(1) I_2O_5 (2) $KI+H_2O$
(3) I_2 (4) $C\ell_2O_6$

CPB327

31. Which of the following statement is not correct?
(1) There are only 12 bonding electron available in one molecule of diborane
(2) $B_3N_3H_6$ is an electron deficient compound
(3) Graphite is thermodynamically most stable crystaline allotrope of carbon
(4) In $Si_2O_7^{6-}$ anion, one oxygen of a SiO_4^{4-} tetrahedron is shared with another SiO_4^{4-} tetrahedron.

CPB328

32. Which of the following group of molecules can act both as oxidant as well as reductant :-
(1) $KMnO_4$, O_3, SO_3
(2) $HC\ell O_4$, HNO_3, H_2O_2
(3) HNO_3, SO_2, O_3
(4) HNO_2, SO_2, H_2O_2

CPB329

33. What is the hybridization on the central atom of SiO_2 ?
(1) sp (2) sp^2 (3) sp^3 (4) sp^3d

CPB330

34. In which of the following reaction, product are not correctly matched ?
(1) $HNO_{3(aq.)} + P_4O_{10(s)} \rightarrow N_2O_3$
(2) $NH_4Cl \xrightarrow{\Delta} NH_3 + HCl$
(3) $Ca_3P_2 + 6H_2O \rightarrow 2PH_3 + 3Ca(OH)_2$
(4) $NH_4Cl_{(aq)} + NaNO_{2(aq.)} \xrightarrow{\Delta} N_2$

CPB331

35. Inert gases are least reactive due to :-
(1) Completely filled valence shell electronic configuration
(2) High ionisation energy
(3) More positive electron gain enthalpy.
(4) All of these

CPB332

36. Which of the following is covalent carbide?
(1) CaC_2 (2) Al_4C_3 (3) SiC (4) Be_2C

CPB333

37. Which of the following halides cannot be hydrolysed?
(1) TeF_6 (2) SF_6 (3) PCl_5 (4) PCl_3

CPB334

38. Which of the following statements is not correct
(1) $FeSO_4$ solution is used for absorption of NO
(2) NH_4NO_3 gives N_2O on heating
(3) $PbNO_3$ gives paramagnetic, brown coloured gas on thermal decomposition.
(4) H_3BO_3 is a Tribasic acid.

CPB335

39. Which product is formed when excess Cl_2 reacts with NH_3 ?
(1) NCl_3 (2) $H_2N–NH_2$
(3) N_2 (4) NH_4Cl

CPB336

40. The point of dissimilarity between $(SO_3)_3$ and $(HPO_3)_3$ (cyclic trimer) is :-
(1) Both have six membered ring.
(2) Both contain central atom in same hybridisation.
(3) Both have equal π bonds
(4) Both are isoelectronic

CPB337

41. Which of following gives only basic product on hydrolysis ?
(1) PCl_5 (2) NCl_3 (3) Mg_3N_2 (4) SO_2

CPB338

42. pπ–dπ bonding is not formed between oxygen and
(1) Phosphours in P_4O_{10}
(2) Nitrogen in N_2O_5
(3) Chlorine in $HClO_4$
(4) Chlorine in anionic part of solid Cl_2O_6

CPB339

43. Which one of the following orders is not in accordance with the property stated against it ?
(1) F > Cl > Br > I; electronegativity
(2) F_2 > Cl_2 > Br_2 > I_2; bond dissociation energy
(3) $F_2 > Cl_2 > Br_2 > I_2$; oxidizing power
(4) HI > HBr > HCl > HF; acidic property in water

CPB340

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44. Ratio of σ & π bond in P_4O_{10} is :-
(1) 4 (2) 3 (3) 2 (4) 1
CPB341

45. Which of the following is correct ?
(1) P–O bonds in P_4O_6 molecule are 6
(2) P–P bonds in P_4 molecule are 6
(3) P–O–P bonds in P_4O_6 molecule are 8
(4) Both (1) and (3)
CPB342

46. Which of the following is not correctly matched ?
(1) XeF_2 and $XeF_4 \Rightarrow$ Non polar but planar.
(2) $XeF_6 \Rightarrow$ exists in solid state as XeF_5^+ and F^-
(3) $XeOF_4 \Rightarrow sp^3d^2$, square pyramidal shape, all identical B.L.
(4) $XeO_3 \Rightarrow$ pyramidal, all bond angles are identical.
CPB343

47. Total number of lone pair of electron and P–O–P linkage present in dimer of P_2O_5 are :-
(1) 16, 4 (2) 20, 6 (3) 22, 6 (4) 30, 4
CPB344

48. Incorrect statement for H_2SO_4 is :-
(1) High volatile
(2) Strong acidic character
(3) Dehydrating agent
(4) Oxidising agent
CPB345

49. Which of the following statement is not correct?
(1) Oxidising power = $SiCl_4 < SnCl_4 < PbCl_4$
(2) The ionic character of lead (II) halide decreases with increase in atomic number of halogen.
(3) Melting point and boiling point of halogen increases down the group due to increase in London-dispersion force.
(4) The liquification of Noble gases decreases down the group.
CPB346

50. Catalyst for Deacon's process :-
(1) $CuCl_2$ (2) Fe_2O_3
(3) V_2O_5 (4) NO
CPB347

51. Which of following species do not give borax bead test ?
(1) Co^{+3} (2) Cr^{+3} (3) Ni^{2+} (4) Au^+
CPB348

52. In layer test of I^- and Br^-. If reddish -brown layer comes first then :-
(1) Br^- present (2) I^- absent
(3) Both (1) and (2) (4) None of these
CPB349

Paragraph for Q. 53 to Q. 56

$$\underset{\text{(Hydrated salt)}}{(A)} \xrightarrow[\text{heated}]{\text{Strongly}} \underset{\text{transparent glassy bead}}{B + C}$$

$$H_3BO_3 \xrightarrow[\text{heated}]{\text{Strongly}} C + D$$

53. Identify C :-
(1) $(BN)_X$ (2) $NaPO_3$
(3) B_2O_3 (4) $Mg(NH_4)PO_4$
CPB350

54. Find the number of water of crystallizations in (A) :-
(1) 4 (2) 5 (3) 10 (4) 24
CPB351

55. How many X–O–X linkages are present in structure of A (X = central atom) :-
(1) 4 (2) 3 (3) 5 (4) 2
CPB352

56. Find the number of tetrahedral and trigonal planar units in structure of A -
(1) 2,1 (2) 2,2 (3) 2,4 (4) 5,2
CPB353

57. In S_8 each sulphur atom is :-
(1) sp hybridised with a planar ring
(2) sp^3 hybridised with a planar ring
(3) sp^3 hybridised with a non-planar ring
(4) sp^3d hybridised two sulphur atoms
CPB354

58. The percentage of p-character in the orbital forming P–P bonds in P_4 is :-
(1) 25 (2) 75 (3) 33 (4) 100
CPB355

59. In order to prepare borazine, we need diborane and ammonia in the ratio
(1) 5 : 9 (2) 1 : 2 (3) 1 : 5 (4) 3 : 4
CPB356

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60. Nitrogen gas is liberated by thermal decomposition of ?
(1) NH_4NO_2 (2) NaN_3
(3) $(NH_4)_2Cr_2O_7$ (4) All

CPB357

61. When excess amount of NH_3 is reacted with Cl_2 gives ?
(1) NH_4Cl (2) N_2
(3) NCl_3 (4) (1) & (2) both

CPB358

62. Most stable tri halides of nitrogen :-
(1) NCl_3 (2) NF_3
(3) NBr_3 (4) NI_3

CPB359

63. The hybridisation of Xe and Sb in product when XeF_4 react with SbF_5 :-
(1) sp^3d, sp^3d^2 (2) sp^3d^2, sp^3d
(3) sp^3, sp^3d^2 (4) sp^3d^3, sp^3d

CPB360

64. $PH_4I + KOH \longrightarrow \underset{\text{(gas)}}{A} + \underset{\text{(salt)}}{B} + H_2O$
Incorrect statement :-
(1) 'A' is inflammable gas in pure form
(2) 'B' is salt of (SB + SA)
(3) PH_3 has bond angle approx 90°
(4) 'A' is a weaker lewis base than NH_3

CPB361

65. In which of the following molecule, vacant orbital is not necessary for dimer formation?
(1) BH_3 (2) $AlCl_3$
(3) CH_3COOH (4) $BeCl_2(s)$

CPB362

66. Stability of monovalent cations of boron family is in the order :-
(1) $Ga^+ > In^+ > Tl^+$ (2) $Ga^+ < Tl^+ < In^+$
(3) $Ga^+ < In^+ < Tl^+$ (4) $Ga^+ > Tl^+ > In^+$

CPB363

67. Which of the following is correct order of backbonding ?
(1) $BCl_3 > BF_3 > BI_3 > BBr_3$
(2) $BF_3 > BCl_3 > BBr_3 > BI_3$
(3) $BF_3 > BCl_3 > BI_3 > BBr_3$
(4) $BI_3 > BBr_3 > BCl_3 > BF_3$

CPB364

68. In the following reactions
$2X + B_2H_6 \rightarrow [BH_2(X)_2]^+ [BH_4]^-$
The amine X will not be -
(1) NH_3 (2) CH_3NH_2
(3) $(CH_3)_2NH$ (4) $(CH_3)_3N$

CPB365

69. Solid CO_2 is :-
(1) Ionic solid (2) Covalent solid
(3) Molecular solid (4) Metallic solid

CPB366

70. Which one of the following statement regarding BF_3 is not correct?
(1) it is a Lewis acid
(2) it is an ionic compound
(3) it is an electron deficient compound
(4) it forms adducts

CPB367

71. Borazole ($B_3N_3H_6$), is related to benzene as:-
(1) isostructural
(2) isoelectronic
(3) both (1) and (2)
(4) both have equal reactivity

CPB368

72. $Na_2B_4O_7.10H_2O$ is correctly represented by:-
(1) $2NaBO_2.Na_2B_2O_3.10H_2O$
(2) $Na_2[B_4O_5(OH)_4].8H_2O$
(3) $Na_2[B_4(H_2O)_4O_7].6H_2O$
(4) None of these

CPB369

73. The hybridization states of boron and nitrogen in borazole are respectively :-
(1) sp, sp (2) sp, sp^2
(3) sp^2, sp^2 (4) sp^2, sp^3

CPB370

74. Alumina is an ionic compound but insoluble in water because :-
(1) its lattice energy is very high
(2) its hydration energy is very high
(3) both lattice energy & hydration energy are same
(4) none of these

CPB371

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75. Carbon forms large number of compounds because it has :-
(1) low electron affinity
(2) variable valency
(3) catenation property
(4) no d-orbitals in valency shell

CPB372

76. Which does not exist ?
(1) $[CF_6]^{2-}$ (2) $[SiF_6]^{2-}$
(3) $[GeF_6]^{2-}$ (4) $[SnF_6]^{2-}$

CPB373

77. Three centred two electron bond is present in:-
(1) NH_3 (2) B_2H_6 (3) BCl_3 (4) $AlCl_3$

CPB374

78. Which of the following order is correct :-
(1) $B_2O_3 > Al_2O_3 > In_2O_3$ (Acidic nature)
(2) $CO_2 > SiO_2$ (Acidic nature)
(3) $CH_4 < SiH_4 < GeH_4 < SnH_4$ (Reducing nature)
(4) All of the above

CPB375

79. Thallium (Tl) Show +1 stable oxidation state because :-
(1) of High reactivity
(2) of inert pair effect
(3) of its amphoteric nature
(4) It is a transition metal

CPB376

80. Maximum no. of atoms lying in a same plane in B_2H_6 are :-
(1) 8 (2) 4 (3) 6 (4) 2

CPB377

81. In which of the following molecule, vacant orbitals do not participate in bonding ?
(1) B_2H_6 (2) Al_2Cl_6
(3) $[H_3N.BF_3]$ (4) Si_2H_6

CPB378

82. Bucky ball or buckminster fullerene is :
(1) An allotrope of carbon
(2) It is referred as C–60
(3) It has sp^2-hybridised nature and resembles with soccer ball
(4) all of these

CPB379

83. Which option is correct ?
(1) Al is more electropositive than Tl
(2) Lewis acidic tendency $\Rightarrow BCl_3 < AlCl_3$
(3) Octet of Boron is complete in BF_3 after back-bonding
(4) $B_3N_3H_6$ has $p\pi$-$d\pi$ type bond

CPB380

84. Which of the following is not correct ?
(1) Borax bead test can be done for Ag^+ ion.
(2) B_2H_6 is colorless gas and highly toxic.
(3) $B_2H_6 + 2CO \rightarrow 2BH_3CO$
(4) CO can acts as lewis base.

CPB381

85. Which of the following compound does not exist ?
(1) SiF_6^{2-} (2) $GeCl_6^{2-}$
(3) $SiCl_6^{2-}$ (4) $[Sn(OH)_6]^{2-}$

CPB382

86. Match the column correctly regarding Borax ?

(a) B–O–B link	(P) 2
(b) B–O	(Q) 5
(c) B-hybridisation-sp^2	(R) 8
(d) Water of crystallisation	(S) 14

	a	**b**	**c**	**d**
(1)	Q	P	S	R
(2)	S	P	R	Q
(3)	Q	S	P	R
(4)	S	R	Q	P

CPB383

87. Which order is correct for degree of Hydrolysis?
(1) $LiCl < BeCl_2$ (2) $BCl_3 < CCl_4$
(3) $BF_3 > BCl_3$ (4) All are correct

CPB384

88. In which of the following dangling bond is not present ?
(1) Fullerene (2) Diamond
(3) Graphite (4) All of the above

CPB385

89. Which is used for measuring high temperature?
(1) B (2) Tl
(3) Ga (4) Si

CPB386

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90. When air is used over hot coke (at 1273 K) instead of steam, a mixture of _____ is produced.
(1) $CO + H_2$
(2) $CO + N_2$
(3) CH_4
(4) $CO + H_2 + CH_4 + CO_2$

CPB387

91. (A) Type of bridge bond in $B_2H_6 \Rightarrow 3C-2e^-$
(B) Maximum replaceable hydrogen in B_2H_6=2
(C) In B_2H_6 number of vacant orbital used by a boron in hybridisation = 2
(D) Number of total regular bonds in B_2H_6 = 6
Correct statements is/are :
(1) A (2) A, C
(3) A, D (4) All

CPB388

92. Stability of cations of group 14 is in the order of :-
(1) $Sn^{+2} < Pb^{+4}$ (2) $C^{+4} < Ge^{+4}$
(3) $Sn^{+2} > Sn^{+4}$ (4) $Sn^{+2} < Pb^{+2}$

CPB389

93. Which of the following halides cannot be hydrolysed?
(1) BCl_3 (2) SiO_2
(3) $BeCl_2$ (4) PCl_3

CPB390

94. H_3BO_3 highly soluble in :-
(1) Hot water
(2) Cold water
(3) Both 1 & 2 remains same
(4) Remains insoluble in both

CPB391

95. Least stable hydride is :-
(1) SnH_4 (2) SiH_4
(3) PbH_4 (4) GeH_4

CPB392

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	1	2	3	1	4	4	4	3	3	3	2	2	2	4	3
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	4	2	4	4	4	4	4	1	3	1	4	3	3	3	1
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	2	4	3	1	4	3	2	4	1	3	3	2	2	1	2
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	3	2	1	4	1	4	3	3	3	3	2	3	2	2	4
Question	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Answer	4	2	1	1	3	3	2	4	3	2	3	2	3	1	3
Question	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Answer	1	2	4	2	3	4	4	1	1	3	3	1	1	3	2
Question	91	92	93	94	95										
Answer	1	4	2	1	3										

d and f Block Elements

1. Which one amongst the following are good oxidizing agents?

A. Sm^{2+} B. Ce^{2+} C. Ce^{4+} D. Tb^{4+}

Choose the most appropriate answer from the options given below :

(1) C only (2) D only

(3) A and B only (4) C and D only

CDF124

2. $K_2Cr_2O_7$ paper acidified with dilute H_2SO_4 turns green when exposed to

(1) Carbon dioxide

(2) Sulphur trioxide

(3) Hydrogen sulphide

(4) Sulphur dioxide

CDF125

3. $KMnO_4$ oxidises I^- in acidic and neutral/faintly alkaline solution, respectively to

(1) I_2 & IO_3^- (2) IO_3^- & I_2

(3) IO_3^- & IO_3^- (4) I_2 & I_2

CDF126

4. Nd^{2+} = _____

(1) $4f^26s^2$ (2) $4f^4$

(3) $4f^3$ (4) $4f^46s^2$

CDF127

5. The correct order of basicity of oxides of vanadium is

(1) $V_2O_3 > V_2O_4 > V_2O_5$

(2) $V_2O_3 > V_2O_5 > V_2O_4$

(3) $V_2O_5 > V_2O_4 > V_2O_3$

(4) $V_2O_4 > V_2O_3 > V_2O_5$

CDF128

6. Which of the following elements have half-filled f-orbitals in their ground state ?

(Given : atomic number

$Sm = 62; Eu = 63; Tb = 65; Gd = 64, Pm = 61$)

A. Sm B. Eu C. Tb D. Gd

E. Pm

Choose the correct answer from the options given below:

(1) B and D only (2) A and E only

(3) A and B only (4) C and D only

CDF129

7. The spin only magnetic moment of $[Mn(H_2O)_6]^{2+}$ complexes is _____B.M. (Nearest integer)

(Given : Atomic no. of Mn is 25)

CDF130

8. When MnO_2 is fused with KOH, a coloured compound is the product and the colour is?

(1) Mn_2O_4, black (2) $KMnO_4$, purple

(3) Mn_2O_3, brown (4) K_2MnO_4, green

CDF131

9. Which of the following species can act as reducing agent ?

(1) $[Co(CO)_4]^-$ (2) $Mn(CO)_6$

(3) $Mn(CO)_5$ (4) $Cr(CO)_6$

CDF132

10. A complex of certain metal has the spin only magnetic moment of 4.91 BM. whereas another complex of the same metal with same oxidation state has zero magnetic moment. The metal Ion could be ?

(1) Co^{2+} (2) Mn^{2+} (3) Fe^{2+} (4) Fe^{3+}

CDF133

11. To an acidified dichromate solution, a pinch of Na_2O_2 is added and shaken. What is observed:

(1) blue colour

(2) Orange colour changing to green

(3) No evolution of oxygen

(4) Bluish - green precipitate

CDF134

12. When a small amount of $KMnO_4$ is added to concentrated H_2SO_4, a green oily compound is obtained which is highly explosive in nature. Compound may be :-

(1) Mn_2O_3 (2) $MnSO_4$

(3) Mn_2O_7 (4) MnO_2

CDF135

13. Chromyl chloride test is given by :-

(1) CH_3Cl (2) AgCl

(3) Hg_2Cl_2 (4) NH_4Cl

CDF136

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14. When $KMnO_4$ reacts with KBr in acidic medium then oxidation state of Mn changes from +7 to ?

(1) +2 (2) +4 (3) +3 (4) +6

CDF137

15. Among the following series of transition metal ion, one where all the metal ion have $3d^2$ configuration ?

(1) $Ti^{+3}, V^{+2}, Cr^{+3}, Mn^{+4}$

(2) $Ti^{+2}, V^{+3}, Cr^{+4}, Mn^{+5}$

(3) $Ti^{+}, V^{+4}, Cr^{+6}, Mn^{+7}$

(4) $Ti^{+}, V^{+3}, Cr^{+2}, Mn^{+3}$

CDF138

16. Which of the following is lanthanide :-

(1) Lanthanum
(2) Californium
(3) Uranium
(4) Europium

CDF139

17. The oxidation state of chromium in the final product formed by the reaction between KI & acidified potassium dichromate solution is ?

(1) +4 (2) +6
(3) +2 (4) +3

CDF140

18. Which of the following ion are Colourless in the aqueaus solution ?

(1) La^{+3} and Lu^{+3}

(2) Nd^{+3} and Pm^{+3}

(3) Ce^{+3} and Pr^{+3}

(4) Sm^{+3} and Eu^{+3}

CDF141

19. Correct order of ionic Radii is :-

(1) $Yb^{+3} < Pm^{+3} < La^{+3} < Ce^{+3}$

(2) $Ce^{+3} < Yb^{+3} < Pm^{+3} < La^{+3}$

(3) $Yb^{+3} < Pm^{+3} < Ce^{+3} < La^{+3}$

(4) $Pm^{+3} < La^{+3} < Ce^{+3} < Yb^{+3}$

CDF142

20. Which of the following ion has highest ionic radius?

(1) Sc^{+3}

(2) La^{+3}

(3) Ce^{+3}

(4) Lu^{+3}

CDF143

21. Which one is least basic ?

(1) $Tb(OH)_3$

(2) $Yb(OH)_3$

(3) $Gd(OH)_3$

(4) $Eu(OH)_3$

CDF144

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	**4**	**4**	**1**	**2**	**1**	**1**	**6**	**4**	**2**	**3**	**1**	**3**	**4**	**1**	**2**
Question	16	17	18	19	20	21									
Answer	**4**	**4**	**1**	**3**	**2**	**2**									

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Coordination Compounds

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1. The primary and secondary valencies of cobalt respectively in $[Co(NH_3)_5Cl]Cl_2$ are:
(1) 3 and 5 (2) 2 and 6
(3) 2 and 8 (4) 3 and 6
CCO229

2. An ammoniacal metal salt solution gives a brilliant red precipitate on addition of dimethylglyoxime. The metal ion is :
(1) Cu^{2+} (2) Co^{2+} (3) Fe^{2+} (4) Ni^{2+}
CCO230

3. The d-electronic configuration of $[CoCl_4]^{2-}$ in tetrahedral crystal field is $e^mt_2^n$. Sum of 'm' and 'number of unpaired electrons is.
(1) 4 (2) 7 (3) 8 (4) 5
CCO231

4. The hybridization and magnetic behaviour of cobalt ion in $[Co(NH_3)_6]^{3+}$ complex, respectively is
(1) sp^3d^2 and diamagnetic
(2) d^2sp^3 and paramagnetic
(3) d^2sp^3 and diamagnetic
(4) sp^3d^2 and paramagnetic
CCO232

5. The number of paramagnetic species from the following is____.
$[Ni(CN)_4]^{2-}$, $[Ni(CO)_4]$, $[NiCl_4]^{2-}$
$[Fe(CN)_6]^{4-}$, $[Cu(NH_3)_4]^{2+}$
$[Fe(CN)_6]^{3-}$ and $[Fe(H_2O)_6]^{2+}$
(1) 1 (2) 2 (3) 3 (4) 4
CCO233

6. Which of the following is correct order of ligand field strength?
(1) $CO < en < NH_3 < C_2O_4^{2-} < S^{2-}$
(2) $S^{2-} < C_2O_4^{2-} < NH_3 < en < CO$
(3) $NH_3 < en < CO < S^{2-} < C_2O_4^{2-}$
(4) $S^{2-} < NH_3 < en < CO < C_2O_4^{2-}$
CCO234

7. Which of the complex has highest paramagnetic nature ?
(1) $[Ni(CN)_4]^{-2}$ (2) $[Fe(H_2O)_6]^{+2}$
(3) $[Ni(CO)_4]$ (3) $[Cu(NH_3)_4]^{+2}$
CCO235

8. Correct order of spin only magnetic moment of the following complex ions is:
(Given At. No. Fe: 26, Co:27)
(1) $[FeF_6]^{3-} > [CoF_6]^{3-} > [Co(C_2O_4)_3]^{3-}$
(2) $[Co(C_2O_4)_3]^{3-} > [CoF_6]^{3-} > [FeF_6]^{3-}$
(3) $[FeF_6]^{3-} > [Co(C_2O_4)_3]^{3-} > [CoF_6]^{3-}$
(4) $[CoF_6]^{3-} > [FeF_6]^{3-} > [Co(C_2O_4)_3]^{3-}$
CCO236

9. The set of correct statements is:
(i) Manganese exhibits +7 oxidation state in its oxide.
(ii) Ruthenium and Osmium exhibit +8 oxidation in their oxides.
(iii) Sc shows +4 oxidation state which is oxidizing in nature.
(iv) Cr shows oxidising nature in +6 oxidation state.
(1) (ii) and (iii) (2) (i), (ii) and (iv)
(3) (i) and (iii) (4) (ii), (iii) and (iv)
CCO237

10. Highest oxidation state of Mn is exhibited in Mn_2O_7. The correct statements about Mn_2O_7 are
(A) Mn is tetrahedrally surrounded by oxygen atoms
(B) Mn is octahedrally surrounded by oxygen atoms
(C) Contains Mn-O-Mn bridge
(D) Contains Mn-Mn bond.
Choose the correct answer from the options given below
(1) A and C only (2) A and D only
(3) B and D only (4) B and C only
CCO238

11. A solution of $FeCl_3$ when treated with $K_4[Fe(CN)_6]$ gives a prussiun blue precipitate due to the formation of
(1) $K[Fe_2(CN)_6]$ (2) $Fe[Fe(CN)_6]$
(3) $Fe_3[Fe(CN)_6]_2$ (4) $Fe_4[Fe(CN)_6]_3$
CCO239

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12. Which of the following are the example of double salt?
(A) $FeSO_4.(NH_4)_2SO_4.6H_2O$
(B) $CuSO_4.4NH_3.H_2O$
(C) $K_2SO_4.Al_2(SO_4)_3.24H_2O$
(D) $Fe(CN)_2.4KCN$
Choose the correct answer.
(1) A and C only (2) A and B only
(3) A, B and D only (4) B and D only
CCO240

13. Which of the following complex will show largest splitting of d-orbitals?
(1) $[Fe(C_2O_4)_3]^{3-}$ (2) $[FeF_6]^{3-}$
(3) $[Fe(CN)_6]^{3-}$ (4) $[Fe(NH_3)_6]^{3+}$
CCO241

14. The complex cation which has two isomers is:
(1) $[Co(H_2O)_6]^{3+}$ (2) $[Co(NH_3)_5Cl]^{2+}$
(3) $[Co(NH_3)_5NO_2]^{2+}$ (4) $[Co(NH_3)_5Cl]^{+}$
CCO242

15. What is the charge on the complex ?
$[Cr(C_2O_4)_2(H_2O)_2]$ formed by Cr(III)
(1) +3 (2) +1 (3) +2 (4) –1
CCO243

16. A complex of platinum, ammonia and chloride produces four ions per molecule in the solution. The structure consistent with the observation is :-
(1) $[Pt(NH_3)_4]Cl_4$ (2) $[Pt(NH_3)_2Cl_4]$
(3) $[Pt(NH_3)_5Cl]Cl_3$ (4) $[Pt(NH_3)_4Cl_2]Cl_2$
CCO244

17. How many moles of AgCl would be obtained, when $Co(NH_3)_5Cl_3$ is treated with excess of $AgNO_3$?
(1) 1 mol (2) 2 mol
(3) 3 mol (4) No ppt is formed
CCO245

18. Which of the following complex can not show isomerism
(1) $[Co(H_2O)_4Cl_2]^{+}$
(2) $[Co(NH_3)_5NO_2]^{+2}$
(3) $[Co(NH_3)_5Cl]^{+2}$
(4) $[Ni(CO)_4]$
CCO246

19. Among the following, metal carbonyls, the C-O bond is strongest in :-
(1) $[Mn(CO)_6]^{+}$ (2) $[Cr(CO)_6]$
(3) $[V(CO)_6]^{-}$ (4) $[Ti(CO)_6]^{2-}$
CCO247

20. Among the following metal carbonyls, in the C-O bond order is lowest in ?
(1) $[Mn(CO)_6]^{+}$ (2) $[Fe(CO)_5]$
(3) $[Cr(CO)_6]$ (4) $[V(CO)_6]^{-}$
CCO248

21. Which of the following are bidentate monovalent anion ligands ?
(a) Acetylacetonato (b) Oxalato ion
(c) Dimethylglyoximato
Select the correct answer :-
(1) (a) only (2) (a) and (c) only
(3) (c) only (4) (b) and (c) only
CCO249

22. Diethylenetriamine is :-
(1) Chelating agent
(2) Tridentate neutral molecule
(3) Tridentate monoanion
(4) (1) and (2) both
CCO250

23. Total number of isomers in $[Pt(NH_3)_2(CN)_2]$ is:-
(1) 2 (2) 4 (3) 6 (4) 8
CCO251

24. Which of the following name is impossible?
(1) Potassium tetrafluoridoxido chromate (VI)
(2) Barium tetrafluoridochromate (II)
(3) Dichlorobis (urea) copper (II)
(4) All are impossible
CCO252

25. The formula of sodium nitroprusside is :-
(1) $Na_4[Fe(CN)_5NO_2]$
(2) $Na_2[Fe(CN)_5NO]$
(3) $NaFe[Fe(CN)_6]$
(4) $Na_2[Fe(CN)_6NO_2]$
CCO253

26. The formula of complex tris (ethylenediamine) cobalt (III) sulphate is :-
(1) $[Co(en)_2SO_4]$ (2) $[Co(en)_3SO_4]$
(3) $[Co(en)_3]_2SO_4$ (4) $[Co(en)_3]_2(SO_4)_3$
CCO254

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27. Which of following pairs of name and formula of complexes, is not correct ?
(1) Tetraamminecopper (II)sulphate..... $[Cu(NH_3)_4]SO_4$
(2) Diamminesilver (I)chloride.. $[Ag(NH_3)_2]Cl$
(3) Potassiumhexacyanidoferrate (III)..... $K_4[Fe(CN)_6]$
(4) Potassiumamminepentachloridoplatinate (IV) ..... $K[PtCl_5(NH_3)]$

CCO255

28. Complex ion $[CuCl_2(NH_2CONH_2)_2]$ is named as:-
(1) dichloro bis (urea) copper (II)
(2) bis (urea) dichloro copper (II)
(3) dichloro bis (urea) copper (I)
(4) bis (urea) dichloro copper (I)

CCO256

29. Trioxalatoaluminate (III) and tetrafluoro-borate(III) ions are :-
(1) $[Al(C_2O_4)_3]$, $[BF_4]^{3-}$
(2) $[Al(C_2O_4)_3]^{3+}$,$[BF_4]^{3+}$
(3) $[Al(C_2O_4)_3]^{3-}$, $[BF_4]^{-}$
(4) $[Al(C_2O_4)_3]^{2-}$, $[BF_4]^{2-}$

CCO257

30. Calculate the EAN of central atom in the following complexes :-
(i) $[Cr(CO)_6]$ (ii) $[Fe(\eta^5\text{-}C_5H_5)_2]$
(1) 29,37 (2) 36,36 (3) 36,27 (4) 27,36

CCO258

31. In sodium nitroprusside, The oxidation number coordination No. and effective atomic number of iron are respectively :-
(1) +3, 6, 36 (2) +2, 6, 36
(3) +3, 3, 36 (4) 6, +3, 35

CCO259

32. In Tollen's reagent $[Ag(NH_3)_2]^{+1}$. The oxidation number coordination number and effective atomic number of central metal ion are respectively :-
[Atomic number of Ag = 47]
(1) +1, 2, 50 (2) +1, 2, 51
(3) +2, 1, 50 (4) +1, 1, 50

CCO260

33. The effective atomic no. of $Co(CO)_4$ is 35 and hence is less stable. It attains stability by :-
(1) oxidation of Co
(2) Reduction of Co
(3) dimerization
(4) (2) & (3) both

CCO261

34. Which of the following shows maximum molar conductance ?
(1) $[Co(NH_3)_6]Cl_3$ (2) $[Co(NH_3)_3Cl_3]$
(3) $[Co(NH_3)_4Cl_2]Cl$ (4) $[Co(NH_3)_5Cl]Cl_2$

CCO262

35. A Coordination complex has the formula $PtCl_4.2KCl$. Electrical conductance measurements indicate the presence of three ion in one formula unit. Treatment with $AgNO_3$ produces no precipitate of AgCl. What is the coordination no. of Pt in this complex :-
(1) 5 (2) 6 (3) 4 (4) 3

CCO263

36. Which of the following complexes produces three moles of Silver chloride when its one mole is treated with excess of Silver nitrate ?
(1) $[Cr(H_2O)_3Cl_3]$
(2) $[Cr(H_2O)_4Cl_2]Cl$
(3) $[Cr(H_2O)_5Cl]Cl_2$
(4) $[Cr(H_2O)_6]Cl_3$

CCO264

37. Concentrated H_2SO_4 will not dehydrate the following complex :-
(1) $[Cr(H_2O)_5Cl]Cl_2.H_2O$
(2) $[Cr(H_2O)_4Cl_2]Cl.2H_2O$
(3) $[Cr(H_2O)_6]Cl_6$
(4) All of these

CCO265

38. On adding $AgNO_3$ solution to a solution of $[Pt(NH_3)_3Cl_3]Cl$. The percentage of total chloride ion precipitated is :-
(1) 100 (2) 75 (3) 50 (4) 25

CCO266

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39. One mole of the complex compound $Co(NH_3)_5Cl_3$ gives 3 moles of ion on dissolution in water. One mole of the same complex reacts with two moles of $AgNO_3$ solution to yield two moles of AgCl(s). The formula of the complex is :-
(1) $[Co(NH_3)_4Cl]Cl_2.NH_3$
(2) $[Co(NH_3)_5Cl].Cl_2$
(3) $[Co(NH_3)_3Cl].2NH_3$
(4) $[Co(NH_3)_4Cl_2]Cl.NH_3$

CCO267

40. Select the correct statement for the complexes $[Cr(H_2O)_6]Cl_3$ and $[Cr(H_2O)_5Cl]Cl_2.H_2O$
(1) They can be differentiated by amount of precipitation.
(2) They can be diffrentiated by electrical conductance measurement method.
(3) They can be differentiated by heating with conc. H_2SO_4.
(4) All of these

CCO268

41. Select incorrect match for $[M(H_2O)_6]^{2+}$ complex :-

	Metal ions	Electronic configuration	FCSE
1	Mn^{2+}	$t_{2g}^3e_g^2$	$0\Delta_0$
2	V^{2+}	$t_{2g}^3e_g^0$	$-1.2\Delta_0$
3	Ni^{2+}	$t_{2g}^6e_g^2$	$-1.6\Delta0$
4	Ti^{2+}	$t_{2g}^2e_g^0$	$-0.8\Delta_0$

CCO269

42. Which of the following is non-conducting?
(1) $CoCl_3.6NH_3$ (2) $CoCl_3.5NH_3$
(3) $CoCl_3.4NH_3$ (4) $CoCl_3.3NH_3$

CCO270

43. Following Sidgwick's rule of EAN, $Fe(CO)_x$ will be :-
(1) x = 4 (2) x = 3 (3) x = 5 (4) x = 6

CCO271

44. Which of the following cannot show linkage Isomerism ?
(1) NO_2^- (2) $SCN^{\ominus}$ (3) $CN^{\ominus}$ (4) NH_3

CCO272

45. The complexes $[Pt(NH_3)_4][PtCl_6]$ and $[Pt(NH_3)_4Cl_2]$ $[PtCl_4]$ are :-
(1) Linkage isomers
(2) Optical isomers
(3) Co-ordination isomers
(4) Ionisation isomers

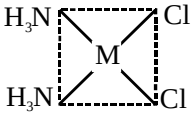
CCO273

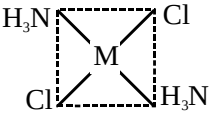
46. $[Co(NH_3)_5NO_2]$ Cl_2 and $[Co(NH_3)_5ONO]Cl_2$ are related to each other as :-
(1) Geometrical isomers
(2) Linkage isomers
(3) Coordination isomers
(4) ionisation isomers

CCO274

47. The no. of geometrical isomers of $[Co(NH_3)_3(NO_3)_3]$ are :-
(1) 0 (2) 2 (3) 3 (4) 4

CCO275

48. Which of following isomers of $[M(NH_3)_2Cl_2]$ would react with silver oxalate $Ag_2(C_2O_4)$?
(1) 

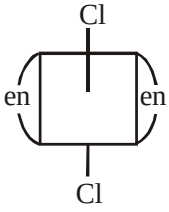
(2) 

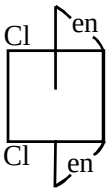
(3) Both (4) None

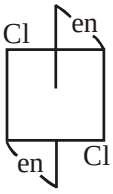
CCO276

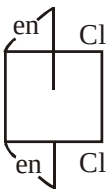
49. Which of the following complex will follow EAN rule ?
(1) $[Mn(CO)_5]$ (2) $[Mn(CO)_6]$
(3) $[Fe(CO)_5]$ (4) $[Co(CO)_4]$

CCO277

50. Identify the geometrical isomers of the following:-


(I)


(II)


(III)


(iv)
(1) I with III (2) II with IV
(3) I with II (4) None of these

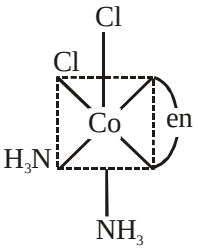
CCO278

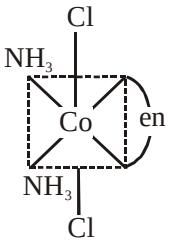
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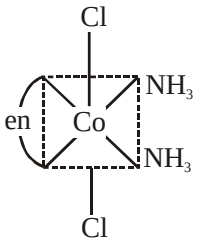
51. Which of the following isomerism is/are not shown by the complex $[CoCl_2(OH)_2(NH_3)_2]Br$?
(1) Ionization (2) Linkage
(3) Gemetrical (4) Optical

CCO279

52. Three arrangements are shown for the complex $[Co(en)(NH_3)_2Cl_2]^+$ pick up the correct statement:-

(I) 

(II) 

(III) 

(1) II and III are optical isomers
(2) I & III are optical isomers
(3) I & II are geometrical isomers
(4) II & III are geometrical isomers

CCO280

53. Which kind of isomerism is exhibited by octahedral $[Co(NH_3)_4Br_2]Cl$?
(1) Geometrical & ionization
(2) Geometrical & optical
(3) Optical & ionization
(4) Geometrical only

CCO281

54. Which of the following has an optical isomer?
(1) $[Co(en)(NH_3)_2]^{2+}$ (2) $[Co(H_2O)_4(en)]^{3+}$
(3) $[Co(en)_2(NH_3)_2]^{3+}$ (4) $[Co(NH_3)_3Cl]^+$

CCO282

55. Which of the following pairs represents linkage isomers ?
(1) $[Pd(PPh_3)_2(NCS)_2]$ and $[Pd(PPh_3)_2(SCN)_2]$
(2) $[Co(NH_3)_5NO_3]SO_4$ and $[Co(NH_3)_5SO_4]NO_3$
(3) $[PtCl_2(NH_3)_4]Br_2$ and $[PtBr_2(NH_3)_4]Cl_2$
(4) $[Cu(NH_3)_4][PtCl_4]$ and $[Pt(NH_3)_4][CuCl_4]$

CCO283

56. Which of the following complexes show geometrical as well as optical isomerism ?
(i) $[Cr(ox)_3]^{3-}$ (ii) $[Rh(en)_2Cl_2]^+$
(iii) $[Co(NH_3)_2Cl_2(en)]^+$
Select the correct code
(1) (i) only (2) (i) & (ii) only
(3) (ii) & (iii) only (4) (i), (ii), & (iii)

CCO284

57. The total No. of possible isomers of the compound $[Cu^{II}(NH_3)_4][Pt^{II}Cl_4]$ are :-
(1) 3 (2) 5 (3) 4 (4) 6

CCO285

58. The number of geometrical isomers for octahedral $[Co(NH_3)_2Cl_4]^-$, square planar $[AuCl_2Br_2]^-$ respectvely are :-
(1) 4, 2 (2) 2, 2 (3) 3, 2 (4) 2, 3

CCO286

59. How many isomers are possible for the complex ion $[Cr(NH_3)_3Cl_3]$?
(1) 3 (2) 2 (3) 4 (4) 5

CCO287

60. Complex
(a) $Na_2[Fe(CN)_5NO]$
(b) $[Fe(H_2O)_5NO]SO_4$
(c) $K_4[Fe(CN)_6]$
Which of the following statement are wrong?
(1) (a) is used in detection of S^{2-}
(2) complex part of each compound is octahedral
(3) (c) is used in detection of Fe^{3+}
(4) Oxidation state of Fe in (a) and (b) is +1

CCO288

61. $[Pt(NH_3)(NH_2OH)(NO_2)(Py)]^+$ will form how many geometrical isomers :-
(1) 2 (2) 3 (3) 0 (4) 5

CCO289

62. Isomerism exhibited by $[Cr(NH_3)_2(H_2O)_2Cl_2]^+$ are :-
(1) Ionisation, optical
(2) Hydrate, optical
(3) Geometrical, optical
(4) Coordinate, geometrical

CCO290

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63. Which of the following is true for the complex $Co(NO_2)(Cl)_2.5NH_3$ (Co is in + 3 oxidation state)
(1) It shows ionisation isomerism
(2) It forms inner orbital complex
(3) It is diamagnetic
(4) All

CCO291

64. The total No. of isomers possible for the complex $[Co(en)_2Cl_2]$ is :-
(1) 2 (2) 3 (3) 4 (4) 5

CCO292

65. The complex $[Pt(NH_3)_4]^{2+}$ has structure :-
(1) Square planer (2) Tetrahedral
(3) Pyramidal (4) Pentagonal

CCO293

66. In complexes most stability is shown by:-
(1) $[Fe(H_2O)_6]^{3+}$ (2) $[Fe(CN)_6]^{3-}$
(3) $[Fe(C_2O_4)_3]^{3-}$ (4) $[FeCl_6]^{3-}$

CCO294

67. Which of the following is not correctly matched?
(1) $[Fe(CN)_6]^{3-}$ – d^2sp^3 and paramagnetic
(2) $[Fe(CO)_5]$ – dsp^3 and diamagnetic
(3) $[Fe(en)_3]^{3+}$ – sp^3d^2 and paramagnetic
(4) $[ZnCl_2(PPh_3)_2]$ – sp^3 and paramagnetic

CCO295

68. The complex $[Fe(H_2O)_5NO]^{2+}$ is formed in Brown ring test for nitrates select correct statement for the complex ?
(1) Hybridisation of Fe is sp^3d^2
(2) Fe and NO has +1 oxidation state
(3) It has spin only magnetic moment of 3.87 B.M.
(4) All are correct

CCO296

69. Which are the correct statements ?
(1) $[Ag(NH_3)_2]^+$ is linear with sp hybridised Ag^+ion
(2) $NiCl_4^{2-}$, VO_4^{3-} and MnO_4^- have tetrahedral geometry
(3) $[Cu(NH_3)_4]^{2+}$, $[Pt(NH_3)_4]^{2+}$ and $[Ni(CN)_4]^{2-}$ have dsp^2 hybridisation of the metal ion
(4) All the correct

CCO297

70. Which of the following is/are inner orbital complex as well as paramagnetic with magnetic moment equal to 3.8 (approx) ?
(1) $[Co(OX)_3]^{3-}$ (2) $[Cr(NH_3)_6]^{3+}$
(3) $[Fe(H_2O)_5NO]^{2+}$ (4) $[Zn(NH_3)_6]^{2+}$

CCO298

71. $Na_2S + Na_2[Fe(CN)_5NO] \rightarrow Na_4[Fe(CN)_5NOS]$. Oxidation number of Fe in reactant (complex) and product (complex) are :-
(1) 2, 1 (2) 2, 2 (3) 2, 3 (4) 3, 3

CCO299

72. Which of the following are square planar complexes ?
(i) $[AuCl_4]^-$ (ii) $[PtCl_4]^{2-}$
(iii) $[Mn(Br)_4]^{2-}$ (iv) $[Cu(NH_3)_4]^{2+}$
Select the correct answer using the codes given below :-
(1) i and ii only (2) ii and iii only
(3) ii and iv only (4) i, ii, and iv only

CCO300

73. The crystal field-splitting for Cr^{3+} ion in octahedral field changes for $I^\ominus$, H_2O, NH_3, $CN^\ominus$ and the increasing order is :-
(1) $I^\ominus < H_2O < NH_3 < CN^\ominus$
(2) $CN^\ominus < I^- < H_2O < NH_3$
(3) $CN^\ominus < NH_3 < H_2O < I^\ominus$
(4) $NH_3 < H_2O < I^\ominus < CN^\ominus$

CCO301

74. Which of the following is a high spin complex?
(1) $[Co(NH_3)_6]^{3+}$ (2) $[Fe(CN)_6]^{4-}$
(3) $[Ni(CN)_4]^{2-}$ (4) $[FeF_6]^{3-}$

CCO302

75. Which has maximum paramagnetic nature?
(1) $[Cu(H_2O)_4]^{2+}$ (2) $[Cu(NH_3)_4]^{2+}$
(3) $[Mn(H_2O)_6]^{2+}$ (4) $[Fe(CN)_6]^{4-}$

CCO303

76. Which of the following order of CFSE is incorrect?
(1) $[Cr(NO_2)_6]^{3-} > [Cr(NH_3)_6]^{3+} > [Cr(H_2O)_6]^{3+}$
(2) $[PtF_4]^{2-} > [PdF_4]^{2-} > [NiF_4]^{2-}$
(3) $[Ni(DMG)_2] < [Ni(en)_2]^{2+}$
(4) $[Co(EDTA)]^- > [Co(en)_3]^{3+}$

CCO304

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77. Which of the following is/are inner orbital complex as well as diamagnetic in nature ?
[Atomic no. → Cr=24, Mn=25, Fe=26, Co=27]
(1) $[Zn(NH_3)_6]^{2+}$ (2) $[Ni(NH_3)_6]^{2+}$
(3) $[Cr(NH_3)_6]^{3+}$ (4) $[Co(NH_3)_6]^{3+}$

CCO305

78. Which of the following is/are correct about $[Cu(NH_3)_4]SO_4$?
(1) It is a Tetrahedral complex
(2) It is paramagentic with one unpaired electron in the d-subshell
(3) It gives white precipitate with $BaCl_2$
(4) Its aqueous solution does not conduct electricity.

CCO306

79. The hybridisation of $[CoF_6]^{3-}$ & $[Co(C_2O_4)_3]^{3-}$ are:-
(1) Both sp^3d^2
(2) Both d^2sp^3
(3) sp^3d^2 and d^2sp^3
(4) d^2sp^3 and sp^3d^2

CCO307

80. $[Sc(H_2O)_6]^{3+}$ ion is :-
(1) Coloured and paramagnetic
(2) Colourless and paramagnetic
(3) Colourless and diamagnetic
(4) Coloured & diamagnetic

CCO308

81. Which of the following complex compounds does not exhibits cis-trans isomerism ?
(1) $[PtCl_2(NH_3)_2]$
(2) $[PdCl_2BrI]$
(3) $[Pt(NH_3)_2(Cl)(Br)]$
(4) $[Pt(NH_3)_3(Cl)]^+$

CCO309

82. Which of the following is correct?
(1) $Ti(NO_3)_4$ is coloured
(2) $[Cr(NH_3)_6]Cl_3$ is colourless
(3) $K_3[VF_6]$ is colourless
(4) $[Ni(CN)_4]^{-2}$ is colourless

CCO310

83. Which is not a π-bonded organo metallic complex ?
(1) Zeise's salt
(2) ferrocene
(3) bis(benzene) chromium
(4) Tetraethyl lead

CCO311

84. What is wrong about the compound ?
$K[Pt(\eta^2-C_2H_4)Cl_3]$
(1) It is called zeise's salt
(2) It is π bonded complex
(3) Oxidation number of Pt is + 4
(4) Four ligands surrounds the platinum atom

CCO312

85. Which of the following order is not correct?
(1) $[Cr(NH_3)_6]^{+3} > [Mn(CN)_6]^{-3} > [V(CO)_6]^-$ spin only magnetic moment
(2) $[Co(CN)_6]^{-3} > [Co(NH_3)_6]^{+3} > [Co(H_2O)_6]^{+3}$ Δ_0 value
(3) $[Ni(CO)_4] > [Co(CO)_4]^- > [Fe(CO)_4]^{-2}$ (M–C bond strength)
(4) $[Ni(NH_3)_4]^{+2} < [Ni(en)(NH_3)_2]^{+2} < [Ni(en)_2]^{+2}$ (Thermodynamic stability)

CCO313

86. Which of the following is correct for zeise's salt ?
(1) The complex ion is square planar
(2) The CMI, Pt is in + 2 oxidation state
(3) $H_2C=CH_2$ Molecule is perpendicular to the $PtCl_3$ plane
(4) All of these

CCO314

87. Which of the following aquated metal ions has the highest paramagnetism ?
(1) $[Cr(H_2O)_6]^{3+}$ (2) $[Fe(H_2O)_6]^{2+}$
(3) $[Cu(H_2O)_6]^{3+}$ (4) $[Zn(H_2O)_2]^{2+}$

CCO315

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88. Coordination isomerism is possible in :-

(1) $[Ag(NH_3)_2][Ag(CN)_2]$

(2) $[Pt(en)_2][PtCl_4]$

(3) $[Pt(NH_3)_4][Pt(C_2O_4)_2]$

(4) $[Pt(NH_3)_4][PtCl_6]$

CCO316

89. Which amongst the following metal carbonyls are inner orbital complex with diamagnetic property ?

(i) $Ni(CO)_4$ (ii) $Fe(CO)_5$

(iii) $V(CO)_6$ (iv) $Cr(CO)_6$

Select the correct code :-

(1) I & II (2) II, III & IV

(3) II & IV (3) I, II, & IV

CCO317

90. Crystal field stablisation energy for high spin d^4 octahedral complex is :-

(1) $-1.8\ \Delta_0$ (2) $-0.6\ \Delta_0$

(3) $-1.2\ \Delta_0$ (4) $-1.6 + P$

CCO318

91. Which of the following is an oxidising agent?

(1) $[Mn_2(CO)_{10}]$

(2) $[Fe(CO)_5]$

(3) $[Mn(CO)_5]$

(4) $[Fe_2(CO)_9]$

CCO319

92. Species which represent maximum stability?

(1) $[Co(CN)_6]^{-3}$ (2) $[Co(C_2O_4)_3]^{-3}$

(3) $[Ni(H_2O)_6]^{+2}$ (4) Both (1) and (2)

CCO320

93. What is the crystal field stablisation energy value of d^5 configuration for tetrahedral complex ?

(1) $0.4\Delta_t$ (2) 0

(3) $-\frac{2}{5}\Delta_t$ (4) $\frac{12}{5}\Delta_t$

CCO321

94. Which of the following complex show maximum numbers of stereo isomers is ?

(1) $\left[PtCl_3(C_2H_4)\right]^{\ominus}$

(2) $\left[CuBr_2Cl_2\right]^{-2}$

(3) $\left[Cr(NH_3)_2(en)_2\right]^{+3}$

(4) $\left[Co(C_2O_4)_3\right]^{-3}$

CCO322

95. A 'Pt' complex of NH_3 and chlorine produces four ions per molecule in the solution is :-

(1) $[Pt(NH_3)_4Cl_2]Cl_2$

(2) $[Pt(NH_3)_6]Cl_4$

(3) $[Pt(NH_3)_2]Cl_4$

(4) $[Pt(NH_3)_5Cl]Cl_3$

CCO323

96. The most stable complex ion is :-

(1) $[Fe(H_2O)_6]^{+3}$

(2) $[Cu(NH_3)_4]^{+2}$

(3) $[Fe(C_2O_4)_3]^{-3}$

(4) $[Ni(dmg)_2]$

CCO324

97. Which of the following complex is example of strongest reducing agent ?

(1) $[Co(CN)_6]^{-3}$

(2) $[Co(CN)_6]^{-4}$

(3) $[Ag(CN)_2]^{-1}$

(4) $[Cu(CN)_4]^{-3}$

CCO325

98. Pair of compound which is planar :-

(1) $[Ni(CN)_4]^{-4}$, $[PtCl_4]^{-2}$

(2) $[NiCl_4]^{-4}$, $[Ni(CN)_4]^{-2}$

(3) $[PtCl_4]^{-2}$, $[Ni(CO)_4]$

(4) $[Ni(CN)_4]^{-2}$, $[Rh(CO)_2(PPh_3)_2]^{+}$

CCO326

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	**4**	**4**	**2**	**3**	**4**	**2**	**2**	**1**	**2**	**1**	**4**	**1**	**3**	**3**	**4**
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	**3**	**2**	**3**	**1**	**4**	**2**	**4**	**3**	**1**	**2**	**4**	**3**	**1**	**3**	**2**
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	**2**	**1**	**4**	**1**	**2**	**4**	**3**	**4**	**2**	**4**	**3**	**4**	**3**	**4**	**3**
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	**2**	**2**	**1**	**3**	**3**	**2**	**3**	**1**	**3**	**1**	**3**	**3**	**2**	**2**	**4**
Question	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Answer	**2**	**3**	**4**	**2**	**1**	**2**	**3**	**4**	**4**	**2**	**2**	**4**	**1**	**4**	**3**
Question	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Answer	**3**	**4**	**3**	**3**	**3**	**4**	**4**	**4**	**3**	**3**	**4**	**2**	**4**	**3**	**2**
Question	91	92	93	94	95	96	97	98							
Answer	**3**	**1**	**2**	**3**	**4**	**4**	**2**	**4**							